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DETECTION OF FALLEN ELECTRIC CABLESON ROAD AND STREETS IN INNER CITIES FOR AUTONOMOUS VEHICLES AND BIKES

UBRAN FALLEN CABLE DETECTION DATASET (UFCDD): A VISUAL BENCHMARK FOR SAMRT MOBILITY IN PAKISTAN

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By

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Mirpur University of Science and Technology (MUST), Mirpur (AJK) Pakistan

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DEDICATION

This work is dedicated to my family, especially my parents, wife, and my beloved kids for continuous support and prayers.

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Humza Nazeer

ABSTRACT*

Abstract

Autonomous vehicles (AVs) need powerful perception systems to function within a complex urban environment safely. Although the modern perception frameworks are effective to the normal road entities, they are still prone to the High-Stakes, Low-Frequency (HSLF) risks, particularly, the thin and irregular objects such as the fallen communication and power cables. Such dangers comprising electrical failure, damage to vehicle and potential loss of control are especially prevalent in the unorganized urban environment of the developing regions where upkeep of the infrastructure is unpredictable. The existing autonomous driving datasets and detection algorithms largely overlook such risks, and as a result, they are poorly generalized and have a high number of false negatives. To address this weakness of fundamental safety provisions, this research paper presents a particular instance of a deep learning-based falling cable danger detection system. At the beginning, a new dataset is designed named Urban Fallen Cable Detection Dataset (UFCDD). It is composed of over 2,500 carefully annotated photographs in different lighting conditions and miscellaneous weather scenarios in the real urban Pakistan locations. Second, to achieve a customized object detection model, based on Convolutional Neural Network (CNN), is used to develop a real-time object detector that can identify thin, curvilinear objects in a chaotic environment. The framework prioritizes precise localization to enable downstream navigation and path planning and classifies cable conditions as being safe or unsafe. According to vast experimentation, the proposed approach achieves an average of 96% at the mean average precision of mAP50 and 76% at the more rigorous mAP50-95 indicating its effectiveness in low-aspect-ratio danger detection and fine-grained localization. These results confirm that correct thin-structure detection is viable in safety-critical autonomous driving conditions and significantly better than generic perception pipelines. Altogether, the article provides a new dataset, a proven real-time detection system, and empirical data on the benefits of safer and more scalable AVs, particularly in developing countries with unorganized and infrastructure-poor urban areas.

Chapter No. 01

INTRODUCTION

1.1 Background

1.1.1 **The Advancement of Autonomous Mobility and Perception Systems**

The invention of the self-driving vehicle (AV), which is currently considered to be one of the most important elements of the modern transport systems, is the result of a dedicated attempt made by the researchers over the last few years to enhance the technologies making the work faster and enhancing the comfort of people. Autonomous vehicles are expected to make mobility, efficiency and public safety undergo substantial improvements and they are classified into 6 degrees of automation (SAE Levels 0 5). The urgency of this kind of technology is emphasized by strong safety statistics: World Health Organization reports show that human error is the main causal factor in the overwhelming majority of traffic accidents that claim the lives of more than 1.35 million people every year in the world. AVs have the potential to considerably decrease these devastating accidents by reducing the impact of human factors, e.g. fatigue, distraction, or impaired driving. Further, AVs can act to ensure environmental sustainability by enabling a faster circulation of traffic and lowering emissions, as well as be more inclusive by giving people with non-communicable mobility the chance to travel by car.

The perception-based system that an AV is required to perform is that of detecting, classifying, and locating objects in its immediate environment, thus making it an indispensable part of its integration. This advanced capability relies on sensor fusion, the combination of data of several modalities. Cameras provide visual and semantic data about color, texture, and lane markings; LiDAR offers high-quality three-dimensional point clouds, which are needed to sense depth and space, whereas radar systems are able to give strong distance and speed data, especially in bad weather when the optical sensors are impaired. The most advanced deep-learning frameworks that are currently used to process these inputs are Convolutional Neural Networks (CNNs) and CNN-Transformer hybrids, which automatically extract hierarchical features of raw data to detect an object (i.e., a vehicle, pedestrian, or traffic sign).

The current perception systems have realised impressive advancement especially in detecting traditional, organised road features. However, they perform poorly when they face geometrically complicated/irregular/non-standard hazards. The design efficiency of current sensorfusion architectures is also a major problem: sensor can afford redundancy with large objects - radar makes up failures in cameras in fog - but sensors have been shown to be deficient with smaller structures. LiDAR frequently provides sparse a point-cloud returns on thin objects and radar cross-sections are so small that they can be discerned. The perception model, in its turn, has to be based largely on high-resolution monocular vision to offset the lack of reliable complementary depth information and place a heavy burden on the detection algorithm on the requirements of extreme accuracy and resistance to background clutter. The fact that systems must have resilience to unexpected environments and infrastructural failures is thus a critical requirement to the introduction of AVs globally, driving specific inquiry into these neglected hazard areas.

1.1.2 **The High-Stakes, Low-Frequency Hazard of Fallen Cables**

Fallen electrical and communication cables on roadways make up a critical type of high-stakes and low-frequency (HSLF) hazards. The inability to identify these impediments can lead to serious mechanical damages of the undercarriage of a vehicle, the total breakdown of a system caused by electrical short-circuit, and even the loss of control that will threaten the population and the occupants of the vehicle (see reference[5]). The high risk of failure in infrastructure due to cables is illustrated by the number of fatalities caused by the power lines accidents in various transport industries; in a case example, the Federal Aviation Authority recorded a 30 per cent rate of aircrafts colliding with power lines between 1997 and 2010.

The issue is further aggravated in unplanned urban settings especially in developing nations where there is high population in-crowding as well as the lack of regular infrastructure maintenance. The literature specifically refers to Pakistani settings, pointing out that these settings pose operational challenges that are barely taken into account through the global standards that are predominantly Western-driven (see reference [6]). The prevalence of electrical lines cutting and landing on roads is also stimulated by the environmental conditions, especially the increased rate of extreme weather conditions, including dust storms, due to climate change (see reference [7]). It requires autonomous vehicles to show consistent functionality in the event of these regionally unique

infrastructure and environmental malfunctions to gain global safety approval and mass deployment.

1.2 Problem Statement

The lack of dependable and real-time detection methods of fallen electric cables in highly congested urban environments destroys the autonomy and scalability of autonomous vehicles technology as a whole.

Low-hanging and fallen electric cables pose a major danger, and this is a road barrier that may affect traffic, destroy complex vehicle functions, and put both passengers and the electric systems of the vehicle at the risk of electrocution [5]. These objects are intrinsically complex and are associated with failures in current perception apparatus; they are inaccessible to generic vision apparatus especially in the context of the complex and changing visual environment of inner-city environments [7]. These cables often occupy very few pixels in a camera image, and because their aspect ratio is very low they cause a visual ambiguity which makes them appear to be menacingly similar to the prosaic artefacts of the background image noise or shadow effects. The false-negative phenomena, therefore, are significantly high, and the risk is completely ignored.

This technological shortage is increased by the lack of training materials. The problem of irregularly shaped and thin threats like electric cables falling is overly ignored in the standard autonomous vehicle datasets, which focus on classic traffic-related data, including pedestrians, cars, and road signs [8]. Such a lack eliminates proper training and assessment of specialized deep-learning models that can detect fine-grained, curvilinear features of such objects. In such a way, a significant safety gap arises because of the absence of special data and specific detection models of thin, irregular hazard.

In order to identify these risks properly and achieve a safe autonomous vehicle operation in the jumbled, unstructured urban setting, the problem is to create a powerful, deep-learning-based detection system supported by a dedicated dataset, called Urban Fallen Cable Detection Dataset (UFCDD).

1.3 Research Questions

The following research questions are aimed at the discussion of the problems outlined in the Problem Statement through guiding the systematic development of the necessary data and detection framework.

RQ1: Data Generation and Fidelity: How can an ad-hoc city be systematically collected and labelled with a specialized, high-fidelity visual dataset (UFCDD) to effectively sample the many geometric and photometric features of fallen and low-lying electric cables in a range of environmental (clear, overcast) and lighting (dawn, dusk, day) conditions? The question helps highlight the basic need to create a dataset that captures the actual complexity and uncertainty of cable hazards in the domain of interest in a satisfactory manner [1].

RQ2: Model Architecture and Optimization: Is it possible to use the customizable Urban Fallen Cable Identification Dataset (UFCDD) to effectively train a Convolutional Neural Network (CNN)-based object identification system, namely, the YOLOv5 architecture, to detect thin, curvilinear cable hazards in real-time and with high accuracy? This question analyses the ability of CNNs, which are known to be feature-extraction networks, to contextualize the technical problem of thin structures. Since the autonomous vehicle systems need low latency to make safety-related decisions, the choice of YOLOv5, a state-of-the-art single-stage detector, is implicitly crucial [2]. The rationale behind this is that YOLOv5 provides the necessary speed and accuracy ratio needed to convert detection to realistic, real-time safety interventions.

RQ3: Performance Verification and Localization Precision: How well can the specialized YOLOv5 framework perform in real time (in terms of measures like mAP50 and the more demanding mAP5095) when trying to identify and classify fallen cable hazards, which proves that the specialized framework has significantly improved robustness to the geometric complexity of thin structures? This focus on strict empirical validation can be explained by the fact that it is very difficult to localize thin objects accurately; hence, rigorous measures like mean Average Precision over Intersection over Union thresholds between 50 -95 are necessary to ensure not only the detection but the accurate delineation of the boundaries, which is crucial to the efficient path planning in the vicinity of hazards. The ability of the model to overcome the geometric ambiguity of thin, low count pixel-count objects is confirmed by accuracy in localization.

RQ4: Safety and Scalability Implication: What does the risk of autonomous vehicles have on scaling smart mobility solutions in developing countries where these risks are common, and how can the proposed framework of detection, made possible by the UFCDD, help to improve the safety and reliability of autonomous vehicles in complex urban settings? Using the technical products or outputs as a means of connecting it with the societal display of goodness, this question is where the research becomes practical in its application and is all-inclusive, especially in situations where the quality of the infrastructures is internationally representative [6].

1.4 Research Contributions

The research adds a number of specialised developments to the current state of the art of the autonomous vehicle perception and safety, especially on the unstructured environment uncommon hazards.

1.4.1.1 Development and Distribution of a new Set of Data.

The main product of this work is the Urban Fallen Cable Detection Dataset (UFCDD) that was created and maintained. This dataset, which is mostly comprised of photographs identified in Mirpur, AJK, Pakistan consists of a rare collection of more than 2,500 carefully annotated photographs of fallen and low-hanging electric and communication lines in urban environments. It fills a serious gap in the available data, which has historically been biased in favor of vehicles, pedestrians, and traffic signs at the expense of cable-related hazards [6]. In order to ensure the resistance to fluctuations in illumination and weather conditions, the UFCDD uses photographs taken in a range of time periods, such as early morning, afternoon, evening; the design is used to eliminate frequent performance deteriorations which take place in autonomous vehicles systems. By targeting the regions that the traditional Western-centric scope of testing has not encompassed, the UFCDD offers the geographical and infrastructural heterogeneity required to create autonomous vehicle systems that can operate safely at the global scale.

1.4.2 Specialization of a High-Performance Detection Framework.

One of them is a salient innovation, the adaptation of a specialised deep-learning object detection framework based on the accuracy and efficiency of the YOLOv5 architecture. To identify spatially fine-grained details (edges, curves, and shapes) that are needed to identify thin and

elongated objects against complicated urban clutter, this framework takes advantage of the ability of convolutional neural networks (CNNs) to automatically learn and extract hierarchical feature encodings of raw pixel data. Being a state-of-the-art single-stage detector, YOLOv5 supports a trade-off between both computational speed and detection accuracy, which is essential to satisfy the low-latency inference demands of autonomous driving that requires safety-level criticality [1, 7, 8, 9, 10]. The model is then trained to classify between the two states of the cable, namely, Safe (intact) and Unsafe (fallen or hazardous) cable states, and thus processes complex visual information down to simple signals that can be processed to support the decision-making of the navigation.

1.4.3 Quantitative Evaluation of Strong Thin-Structure Detection.

The present study empirically confirmed the performance of the CNN-based YOLOv5 model, which provides the quantifiable information about the effectiveness of the model in the detection of thin, irregular hazards. It was accurate (mean Average Precision (mAP)) on a range of the IoU thresholds of 0.50 to 0.95 (76 or 96 percent), and 96 percent at an IoU threshold of 0.50. The strong score of mAP 50 means the presence of strong classification performance, which is always able to separate hazardous wires and background imagery. Besides, the 76% accuracy on the mAP at the 50-95 threshold point is worth mentioning. These findings show the framework has defeated the inherent geometric difficulty of low-count-pixel objects, since this measure requires good delineation of boundaries, which is captured in high intersection over union values. This stringent validation support a stable detection system which has the capacity to improve urban traffic safety by proving the ability of the model to pick out thin and irregular cable structures even in a difficult urban environment [13].

Table 1.1: Key Contributions and Performance Benchmarks

Contribution Element	Description/Metric	Significance
Dataset Name	Urban Fallen Cable Detection Dataset (UFCDD)	Addresses overlooked High-Stakes, Low-Frequency (HSLF) hazards [14].

Dataset Size/Scope	2,500+ images from Mirpur, AJK, Pakistan	Provides critical geographical and infrastructural diversity for global AV deployment [15].
Detection Framework	CNN-based YOLOv5 Model	Ensures real-time, low-latency performance required for safety-critical AV tasks [16].
Mean Average Precision (mAP@50)	96%	High accuracy for primary detection threshold, indicating strong capability in distinguishing hazardous objects [17].
Mean Average Precision (mAP@50-95)	76%	Robust performance across stricter localization tolerances, confirming precision on thin, complex structures [18].

1.5. Research Objectives

To fill the identified safety gap in a systematic manner by means of data collection and model application, a set of specific objectives is developed to be used as the guiding framework of the research.

1. **Hazard Image Collection:** A large database of photo documentation of collapsed power and communication lines is gathered across several Pakistani cities and hence the data set is a fair reflection of the reality of the infrastructure heterogeneity. The collection is scoped to cover a few Pakistani areas, specifically Mirpur, to consider the differences in infrastructural planning and geographical setting.

2. **Acquisition of Environmental Diversity:** Data are collected within a wide range of natural conditions of light, cloudiness or overcast, and temporal conditions, e.g. dawn, late morning, afternoon, and dusk. This will ensure that the variation of visuality in dynamic situations is added to enhance the strength of the model training process.
3. **Manual Annotation and Ground Truth Generation:** To produce a robust ground truth, the gathered imagery is annotated with a special care by skilled annotators, and special attention is paid to the correct determination of the spatial boundaries of fallen cables. The annotation protocol is critical in offering supervision that is needed to train the deep learning model subsequently.
4. **Construction of Dataset:** The Urban Fallen Cable Detection Dataset (UFCDD) is carefully prepared, checked and published as a publicly available material. The primary application of this dataset is to verify and train hazard detection systems of autonomous vehicles and the general robotics, whether in drone navigation or surveillance.
5. **Model Development and Validation:** A convolutional neural network object detector model, YOLOv5, is trained on the UFCDD and a quantitative evaluation of its performance is provided based on such established measures as mAP50, mAP50-95, precision, recall, and F1-score. These tests verify the appropriateness of the model to low-latency safety-critical autonomous vehicle operations as described in references [11, 12, 13].

1.6. Research Gap

The research chosen on this particular basis is timely because it will fill a variety of notable gaps that will limit the security and dependability of autonomous vehicles, especially in different and unstructured urban areas across the globe.

1.6.1 Geometric and Photometric Intrinsic Detection Problems

Fallen cables have their own peculiar physics and geometry, which form the gist of the technical deficiency, and are a fundamental problem to the traditional computer vision systems. The extremely low aspect ratio of them is frequently represented in camera images as sparse lines with one or a few pixels, making cables virtually invisible or indistinguishable by a generic object detector. The accuracy that is needed to capture or categorize these tiny, sub-scale objects is not available in systems based on traditional bounding-box systems, which are designed to handle the

larger scale, e.g. vehicles. In addition, the visual scene has a lot of photometric ambiguity. Fallen cables generally merge on the background noise, shadows, and non-important linear elements, and have low contrast to the complicated road substrates (concrete, asphalt, dirt). The conventional perception pipelines do not have the capability to handle this problem thus requiring the extraction of fine-grained information like faint edges and curves [18].

1.6.2 Standard Autonomous Vehicle Dataset Deficiencies

In the case of the recognized benchmarks, the prevalence of a lack of data is one of the key factors preventing the creation of a reliable detection system. In this effort, traditional AV datasets, such as major publicly available ones, such as KITTI, NuScenes, and BDD100K, do not suffice. Most of these standards were gathered in very civilized urban or highway environments, which inherently reduce the likelihood of facing unanticipated infrastructure failures like utility failures or large amounts of debris. The gap in the research is characterized by two significant gaps in the available data:

Annotation Scarcity: These popular datasets do not have the required annotation and the special taxonomy to support thin and non-conventional hazards. This lack impedes the creation of machine-learning models that can be generalized to dynamic and highly cluttered situations.

By covering unrelated Debris Focus: whereas tailored datasets (e.g. HazardNet) represent more general road debris, they tend to use height limits specifically disallowing the critically low-lying geometry of fallen electrical line, in favor of larger, non-curvilinear objects, like tires or building materials.

As a result, there is a lack of training data on the specific models, resulting in the use of the characteristics acquired under irrelevant or benign settings as well as sub-optimal detection of high-speed low-friction cable (HSLF) risks. In the UFCD, the direct solution to this data gap is by choosing imagery that is peculiar to high-danger, unstructured urban road scenarios.

1.6.3 The Domain Transfer and Infrastructural Gap

The gap in operation is significant as the research provides by previous researchers and the present necessity of ground-based autonomous vehicles indicates. One of the key subjects of the previous

line-detection literature involved the use of unmanned aerial vehicles (UAVs) to inspect overhead power lines and pylons [10]. Such techniques applied the process of feature grouping, Kalman filtering and Radon transforms that worked well in controlled conditions where the background could be evaluated [11]. However, there are monumental challenges as far as the implementation of these airborne methodologies on ground AVs is concerned. Aerial systems are favored by straight, line geometries, and in high contrast (wire versus clear sky) structures. On the contrary, ground AVs have to deal with a very wide background clutter, complicated road textures, visual interference caused by rain, overly complicated and curved and non-rigid paths of fallen cables. This can be attributed to these topological and environmental differences, which have to be significantly changed in terms of detection algorithms. Moreover, the focus on unorganized urban environments like in Pakistan points to an essential infrastructural and geopolitical lapse in the development of AV safety. High rates of failed cables in these locations are an expected threat linked to infrastructural density, periodic maintenance, and stress factors of the environment, and not a chance occurrence [6]. The need to have a region-specific data set, including the UFCDD, indicates that global AV maturity (SAE Level 5) will need modular perception stacks, and such stacks must be able to reliably adapt to local infrastructures and local hazards of variable quality, as opposed to just relying on centralized, generalized models. This research helps to make the smart mobility solutions that are being implemented in developing countries scalable, as it forms the background upon which the specific geographical shortages can be identified and addressed.

Chapter No. 02: Relevant Research in This Field

Chapter No. 3: Give a thorough description of the materials and experimental setups.

Chapter 4: Outcomes of research and recommended configurations.

Chapter No. 5: How to Expand This Research and Potential Future Directions for Research Extension.

Chapter No. 02

RELATED WORK

2.1. Foundations of Autonomous Vehicle Perception and System Architecture

In order to achieve completely autonomous mobility (SAE Levels 4 and 5), highly dependable perception systems that can achieve comprehensive environmental knowledge under all operational scenarios must be developed.[1] The perception system supplies the crucial, safety-critical data input for the ensuing decision-making processes in autonomous vehicles (AVs), which function on a linked stack of modules that includes perception, localization, planning, and control.[2]

2.1.1. Perception Requirements and Sensor Modalities

In order to overcome the inherent limits of any one type of sensor, robust perception systems in contemporary AVs rely on a multi-modal sensor suite.[3]

The Visual Modality (Cameras) is essential for understanding complicated scenes since it provides rich semantic information, color, texture, and lane markers.[1] To recognize objects in dynamic urban surroundings, camera data is fed into advanced deep learning architectures like You Only Look Once (YOLO) [42] and Faster R-CNN [13].[1] Despite being affordable and easy to use, vision systems are very vulnerable to performance degradation in difficult circumstances, such as low light, strong glare, fog, or rain, which jeopardizes image quality and system dependability.[7]

Ranging modalities, principally Light Detection and Ranging (LiDAR) and Radar complement visual data. LiDAR provides high-quality three-dimensional (3D) point clouds that are accurate spatially, geometrically, and in distance, and can be used to achieve a high degree of localization and obstacle avoidance [1]. By contrast, radar is very efficient with regard to the measurement of speed and distance that enables it to be more resilient in adverse weather conditions (rain, fog, dust) in which optical sensors often fail to operate [4].

2.1.2 Sensor Limitations in Detecting Thin Structures

When such traditional sensor modalities are brought to contact thin, geometrically complex materials, such fallen electric lines the result is a big problem. Wire sensors are unreliable in the fine wire detecting process because of their nature.

The slenderness of the profile of a cable when compared to its length, or its very low aspect ratio, results in a sparse point cloud when LiDAR is used. This absence of geometric information makes LiDAR processing methods poor at determining the obstruction [5].

The vision-based system is disproportionately burdened by ranging sensors' incapacity to accurately detect these dangers.[5] As a result, in situations where geometric sensors are unable to extract the fine semantic properties (edges, shading, texture) of the cable structure, successful identification must heavily rely on high-resolution visual processing systems and specialized deep learning models.[5] The architectural limitations covered in following sections are driven by this required research focus.

2.2 The Critical Challenge: High-Stakes, Low-Frequency (HSLF) Obstacles

Even though the most advanced AV perception systems available today are very good at recognizing common, structured traffic elements, cars, pedestrians, and traffic signs, they are severely limited when it comes to uncommon, geometrically complex, or unusual road hazards, which are collectively referred to as road debris or infrastructure failures.[1]

2.2.1 Defining Unconventional Road Hazards and Infrastructure Failures

One important class of High-Stakes, Low-Frequency (HSLF) dangers is represented by fallen power and telecom cables on metropolitan roads.[1] Running over such an impediment could result in catastrophic consequences, such as serious mechanical damage to the vehicle's undercarriage, system failure from electrical shorts, and loss of control that puts both the public and the occupants of the vehicle in danger.[1] Organizations like the American Automobile Association (AAA) and the Federal Aviation Administration (FAA) have highlighted the

seriousness of collisions and crashes caused by power lines and road debris, confirming the need for reliable detection systems.[17]

This safety concern is worse in informal urban areas such as that of poor countries like Mirpur, Pakistan [25]. These environments increase the risk of the electrical cables falling on the road by having high infrastructure density, uneven maintenance as well as decreasing the resistance to environmental factors especially the rising rate of severe weather conditions such as dust storms which break the cables. It is necessary that systems should be capable of detecting and dealing with these geographically distinct infrastructural deficiencies so that autonomous mobility may have worldwide, scalable implementation [25].

2.2.2 The Geometric Ambiguity Problem

However, in contrast to traditional, massive obstacles, fallen power cables are particularly geometric and photometric problems. Geometrically speaking, they have a very low aspect ratio, so they are very thin against their length. They are often a very small number of pixels (occasionally 1 to 3 pixels) in the high-resolution camera photographs, particularly when observed at a distance [43].

This low pixel count generates a lot of visual ambiguity and bewilderment. Cast shadows, deep road surfaces, redundant background noise, inoffensive linear elements such as painted markers or the crevices of the roads can all easily be confused with thin things,[45]. Given the time of the day or the weather, the visual image of the cable (contrast, texture, colour) concerning the road surface differs greatly, producing low-contrast situations that disorient generalist detectors[45].

Such geometric complexity means that in a high-clutter, unstructured environment, models will suddenly exhibit a sudden drop in performance referred to as domain shift when applied. The rationale behind this is that models that are trained on structured data implicitly learn not to be impacted by a high density of visual clutter.[45]. The model continuously fails catastrophically in generalization after being trained on clean data, completely overlooking the uncommon yet very important HSLF hazard [45].

Table 2.1 Thin-Object, Boundary-Aware, Powerline & Infrastructure Detection Research

Ref No.	Area / Focus	Key Findings	Relevance to Fallen Cable Detection
[9]	Aerial Power Line Extraction	Edge + region-based extraction	**Foundational thin-line detection** basis
[18]	Automated Power Line Inspection (Review)	**DL approaches** for line inspection	Techniques **transferable** to fallen cables
[36]	Power-line Segmentation	Semantic segmentation of overhead lines	**Directly adaptable** for fallen cable segmentation
[44]	Synthetic Wire Detection	**Synthetic → real** training improves robustness	Useful for **rare cable-fall training data**
[46]	Wire Detection + Depth	Cable and **depth estimation**	Useful for **locating** fallen wires on road
[51]	Small-object Sensitive Segmentation	**Attention** boosts thin-object sensitivity	**Critical** for long thin cable structures

[50]	Boundary-aware Loss Functions	Better **segmentation edges**	Helps separate cable pixels cleanly
[58]	**Thin-structure Obstacle Detection**	**Fast detection** of thin rods/wires	**Closest match** to fallen cable detection problem
[59]	Powerline & Pylon Detection	**Robust cable detection** from aerial views	**Cable shape priors** applicable to street-level detection

2.3 Evolution of Hazard Detection Methodologies

The necessity to detect detailed, thin structures on the ground with high accuracy has triggered the technological progress that brought a significant shift in the conventional computer vision to the contemporary deep learning algorithms.

2.3.1 Classical Computer Vision Approaches

The primary theme of the first utility infrastructure studies was the inspection and detection of overhead power wires and pylons with the aid of imagery captured by unmanned aerial vehicles (UAVs) [1]. These conventional methods largely relied on the manual feature extraction, which was mathematically intricate. The Hough and Radon transforms that were utilized to detect and reconstruct straight line segmentation based on predicted geometric saliencies were noteworthy techniques[48]. Such rebuilt line structures were then followed through consecutive frames by other techniques such as Kalman Filtering which aided in inspection[1].

But upon application to the terrestrial realm of self-driven vehicles, these traditional methods have become very delicate and unreliable. They were so much sensitive to shadows, changes in the road texture in Paris, and interference with the background, [1]. Above all they could not guess the multiplied, wavy, and non-rigid paths that, on the land, fallen cables take--a topology quite

the opposite of the straight, structured, tautness of overhead power lines[1]. This lack of ability to generalize in complex ground clutter necessitated the adoption of a stronger methodological paradigm.

2.3.2 The Paradigm Shift to Deep Learning

The popularity of deep learning (DL) methods is because of the nature of classical methodologies, which have certain weaknesses[54]. Convolutional Neural Networks (CNNs) have provided the foundation of trustworthy and effective recognition because they automatically learn a hierarchy of intricate patterns and fine details on raw picture data.[1]. This allowed systems to deal with complex background clutter and the visual difference that occurs with changing lighting and weather conditions, and it also eliminated the intensive task of designing elements manually[1].

The technical challenge of transferring domain of aerial inspection to ground hazard detection is a significant technical barrier in the area of thin structure detection.[1]. Whereas on the ground AV systems need to distinguish between dark cable and textured roads, which are often covered by shadow or wetness, the situation is more advantageous with aerial detection where the difference between the wire and the sky or a clear surface is high (high contrast video) [22]. Due to this complexity, CNNs alone can acquire a dedicated feature set that specializes in ground texture and shadow discrimination.

2.4 Deep Learning Architectures for Real-Time Object Detection in AVs

The safety-critical application of autonomous driving necessitates object detection systems with low latency (real-time inference) on edge computing hardware with limited resources.[1] High-efficiency architectural solutions are chosen due to this demand.

2.4.1 Taxonomy of Modern Detectors and the YOLO Framework

The two broad types of modern object identification algorithms are two-stage (such as Faster R-CNN) and single-stage (such as YOLO, SSD) detectors[1]. Single-stage detectors (particularly the YOLO family (You Only Look Once)) are popular in AV perception (as opposed to two-stage systems) due to their simplicity and their ability to run at the fast inference speed (low latency) required by real-time safety maneuvering.

The YOLOv5 model that was selected to use in the proposed framework is an advanced model that attempts to find a balance between speed and accuracy[1]. Its architecture comprises of a detecting head, multi-scale feature integration (through a neck network) and a robust backbone feature extraction system[1]. Its natural ability to simultaneously evaluate multiple resolutions of input images provides a structural benefit in recognizing small objects compared to single input resolution tuned models[1].

2.4.2 Hybrid Models and Lightweight Backbones for Edge Computing

Architectural frameworks that are more specific than the general hybrid approaches are often required by the natural conflict between low computing load necessary (to achieve real-time deployment) and high precision necessary (to extract minute cable details).[1].

Convolutional networks are updated with high accuracy detection models such as ConvNeXt, which are at the state-of-the-art performing on a par with Vision Transformers [2]. ConvNeXt would be an excellent choice of the high-precision refinement layer required in identification of thin objects due to its ability to extract complex feature maps[59]. Conversely, some models like MobileNetV4 are compact networks that are used to perform low-latency inference and optimize their performance on edge devices [5]. Universal Inverted Bottleneck (UIB) module is an important component that reduces the computation load and enhances the speed [7].

To obtain the highest precision of detection in thin cable identification and maintain the minimal latency necessary to operate safely, the architectural scheme integrates the strengths of ConvNeXt (in terms of robustness in features) and MobileNetV4 (in terms of computational efficiency) [9].

Multi-Scale Feature Fusion is used for additional optimization. When fine features and spatial information are lost during the deep network down sampling process, Small Object Detection (SOD) suffers.[4] In order to mitigate feature distribution discrepancies between resolutions, architectures that use Feature Pyramid Networks (FPNs) within the neck network efficiently integrate low-level fine edge features (for precision) with high-level semantic information (for context).[61] In order to maintain the modest visual properties of a thin cable throughout the detection pipeline and improve localization accuracy, this feature aggregation approach is essential [4].

For detecting continuous abnormalities like long power lines, sophisticated hybrid models that combine CNNs for local feature detection and Transformers (using self-attention for global contextual comprehension) have also shown great success [55].

Table 2.2: Detection Models & Algorithms Relevant to Fallen Cable Detection

Ref No.	Model / Method	Contribution / Insight	Relevance to Fallen Cable Detection
[20]	**HOG + SVM** (Dalal & Triggs, 2005)	Classical gradient-based detection	Useful **baseline** for thin-object edges and cable contours
[21]	**Focal Loss** (Lin et al., 2017)	Handles imbalance and **small objects**	Fallen cables are **rare**, thin, elongated objects
[27]	Keypoint UAV Detection + **Rotated DIoU**	Detects rotated or elongated objects	Captures cable-like **long shapes** on roads
[28]	**LI-YOLO** (Low-light detection)	Strong detection in low illumination	Useful for **night-time** cable detection
[29]	Deep Learning Object Detection Survey	Overview of classical and modern detectors	Helps identify strong **baselines** for cable detection

[30]	**SSD**	Lightweight object detector	Suitable for **bike-mounted hardware**
[31]	**MobileNetV2**	Efficient convolutional backbone	**Real-time detection** on edge devices
[36]	Power-line Segmentation (Damodaran)	Segmentation of overhead lines	Applicable to fallen cable **geometry**
[44]	**Dilated CNN for Wire Detection**	Wire detection using **synthetic data**	Highly relevant for **rare-cable-event synthesis**
[51]	Small-object Sensitive Segmentation	**Across-feature-map attention**	Helps detect **thin cable-like structures**
[58]	**Thin-structure Obstacle Detection**	Detects thin rods/wires	Almost **directly applicable** to fallen wires
[59]	Cable & Pylon Detection	Vision-based cable detection (aircraft)	**Cable shape priors** transferable to road scenarios

[50]	**Boundary-aware Loss**	Improves fine object boundary segmentation	Useful for segmenting **cable edges** on road
[49]	**DL vs Traditional CV**	Comparative analysis	Helps **model-selection** for cable detection
[56]	**ConvNext + YOLOv5**	Strong small-object detection performance	Good for **thin-object detection**
[54]	**Loss Function Review**	Comprehensive loss analysis	Helps pick best loss for **thin-structure segmentation**

2.5 Specialized Techniques for Thin and Curvilinear Structure Perception

Because fallen cables are fundamentally geometric, detection techniques must put pixel-level boundary accuracy ahead of conventional bounding box localization.

2.5.1 Representation and Boundary Accuracy

Traditional bounding box detection systems such as Single Shot Detectors (SSDs), simply cannot properly identify objects with a few pixels because they are physically programmed to look for bigger objects [23]. The result of the bounding box is not accurate enough to enclose or categorize thin structures [24].

Consequently, Semantic Segmentation, a pixel-wise categorization method is regarded as the most suitable method of establishing the precise physical scale of the risk [25]. Semantic segmentation correctly outlines the outline of the item by assigning each pixel in the image a categorical name. It is also a pixel-level granularity that is necessary to allow the vehicle to

accurately execute an avoidance maneuver [26]. The simplest models in this area are the architectures of Fully Convolutional Networks (FCNs) and U-Net [27].

As an alternative to the attempt to split every single pixel, a more specialized approach known as Key-Point Encoding models curvilinear wire as a set of uniformly spaced key points [1, (Liu et al., 2020)]. It simplifies the process of detection and makes it possible to accurately reconstruct the object curve with implicit functions, which are quite common in frameworks designed to estimate posture and other thin objects.

2.5.2 Contextual and Depth Requirements for Safety

The conceptual framework of ground AVs detection task is a hybrid task: the object detection (YOLO) speed is required to be responsive in real-time, whereas the safety of avoidance is contingent upon the pixel-level accuracy obtained by means of segmentation training principles.

In addition, since they are visually indistinguishable between a slender cable and a low-lying shadow/road crack [46], two-dimensional (2D) detection is associated with an inherent false positive/negative risk. The validation in 3D space is required to guarantee the robust detection when it comes to ensuring that the object is a true physical hazard [46]. The development of Joint Detection and Depth Estimation (e.g., UCorr) is essential to ground AVs because they will need 3D data to tell whether a piece of hazardous cable on the road is a hazard or a non-harmful linear marker [59]. Finally, in many cases the implementation of high-confidence safety may have to involve non-visual confirmation like monocular depth estimates or even special sensor inputs (e.g. magnetic field sensing of live wires) to add redundancy and reduce visual uncertainty [60].

2.6 Addressing Data Scarcity and Training Difficulties

Beyond building design, the difficulties in identifying uncommon, thin risks include basic problems with data quantity and quality, particularly with regard to extreme class imbalance (EVI).

2.6.1 The Problem of Extreme Class Imbalance (EVI)

At the pixel level, thin cable detection is essentially an EVI problem. The little number of pixels that constitute the thin wire (the positive class) is dwarfed by the vast majority of pixels that belong to the background (the negative class) in any given metropolitan image [46].

The network is biased toward predicting the dominant background class during training with traditional loss functions (such as Cross-Entropy) due to the large number of background samples [66]. This bias has serious consequences, including poor delineation of the few positive pixels that mark the cable boundary and high rates of false negatives, which cause the key hazard to be ignored [66].

2.6.2 Specialized Loss Functions for Robust Learning

Conventional loss functions must be supplemented or replaced with specialized alternatives that prioritize the few crucial cable pixels in order to lessen the systemic failure brought on by EVI [66].

The contribution of easily categorized negative examples (background) is greatly reduced by Focal Loss (FL), which dynamically re-weights the standard cross-entropy loss [48]. This method significantly increases accuracy in cluttered, unbalanced images by forcing the model's training to concentrate on the uncommon, hard negative examples, which are frequently found close to the object border [48].

For segmentation tasks involving extreme imbalance, Dice Loss and similar region-based losses (such as Tversky Loss and Generalized Dice Loss) are quite successful [66]. The goal of these losses is to maximize the spatial overlap (also known as the Intersection over Union, or IoU) between the ground truth and forecast regions [66]. Furthermore, specific boundary-aware loss functions are frequently required to compel the network to prioritize prediction confidence, particularly near object contours, since the misclassification of boundary pixels seriously impairs the perceived quality and safety utility of thin structure detection [66]. By applying a severe penalty for errors involving the minority (cable) class, the selection of these particular loss functions serves as an algorithmic safety guarantee and mitigates the statistical bias of the network.

2.6.3 Synthetic Data Generation (SDG)

The use of Synthetic Data Generation (SDG) techniques is required due to the practical difficulties in gathering adequate amounts of real-world imagery for HSLF events [26]. SDG is a viable option for unusual object classes, utilizing generative models such as Diffusion Models and Generative Adversarial Networks (GANs) or simulation-based rendering [15].

It has been demonstrated that pre-training models on artificially produced data with significant variation in pose, illumination, and occlusions, followed by fine-tuning on a small sample of real data, improves model resilience and speeds up development [66]. Crucially, the real-world dataset (the Urban Fallen Cable Detection Dataset, UFCDD) offers the environmental factors and realistic visual textures required for the target unstructured urban area (e.g., Mirpur, AJK, Pakistan) [66]. In order to ensure that SDG can provide variability without compromising domain realism and to enable efficient transfer learning, this real-world anchor is crucial for preserving integrity during synthetic scaling [15].

2.7 Analysis of Data Deficiencies and Domain Shift

Fundamental flaws in the training and validation datasets are directly responsible for the generalized AV perception systems' consistent inability to accurately identify falling cables.

2.7.1 Critical Review of Standard AV Benchmarks

KITTI, NuScenes, and BDD100K are globally known AV benchmark datasets that cover activities related to organized traffic environments (e.g., automobiles, pedestrians, lane lines) in highly established urban or highway settings [35].

In terms of thin dangers, these records show a serious systemic deficit [35]. Extremely thin objects that generate sparse returns are frequently systematically filtered out of the ground truth labeling process due to the strict quality assurance requirements during data curation, which frequently require a minimum number of confirming sensor returns (e.g., LiDAR points) to validate an annotation [44]. A widespread safety blind spot for thin, low-profile impediments is unintentionally created by this deliberate selection for traditional, well-defined targets, guaranteeing that models trained solely on this conventional data are specifically educated to avoid these HSLF threats [35].

2.7.2 Existing Datasets for Debris and Unstructured Ground

The particular risk of falling cables is not covered by alternative databases, even when they address other types of environmental complexity.

Unstructured Ground Datasets, like Kamino and the Robot Unstructured Ground Driving (RUGD) dataset, focus on scene semantics and general terrain drivability while addressing the difficulties of off-road or unpaved navigation [50]. These tools are useful for addressing excessive environmental clutter, but they do not provide the detailed, pixel-level annotation and particular taxonomy needed for infrastructure failures such as electric lines [50].

Larger, vehicle-damaging debris, such tires, bricks, or building materials, is given priority in road debris datasets (like HazardNet) [24]. The low-lying, curved infrastructural components that are essential to this study are essentially excluded by the hazard definition in these datasets, which frequently includes a minimum object height requirement (e.g., more than 15 cm) [24].

This investigation validates a distinct and crucial gap in the available public resources: no dataset currently in use offers both the specialized, required annotation taxonomy for fallen electric and communication cables and the necessary visual complexity (unstructured urban areas with thick clutter).

Table 2.3: Datasets Useful for Road Scene Analysis and Cable/Thin Object Detection

Ref No.	Dataset / Source	Description	Relevance to Fallen Cable Detection
[16]	**Cityscapes**	Urban street semantic segmentation	Provides **road context** for cable simulation

[32]	**Synthetic Domain Randomization Dataset**	Synthetic training via domain randomization	Enables training on **rare or dangerous cable-fall scenarios**
[33]	**RUGD Dataset**	Unstructured navigation scenes	Useful for debris and **obstacle robustness**
[37]	Bridge Damage Dataset	High-resolution **crack/line features**	Thin cracks **resemble long cable structures**
[38]	Road Spill Detection Dataset	**Hazard detection** on roads	Fallen wire treated as road hazard
[46]	**UCorr Wire Detection Dataset**	Wire detection + depth for drones	Strong training set for **cable geometry modelling**
[47]	Line Segment Dataset	Line detection benchmarking	Useful for cable **line-segment extraction**
[57]	M4MLF-YOLO Dataset	**High-precision segmentation**	Good for **fine-edge cable segmentation**
[58]	**Thin Obstacle Dataset**	Thin wires/rods in scenes	**Closest dataset** to fallen cable detection

[59]	Cable & Pylon Detection Dataset	Power line imagery (aircraft)	Provides **cable shape and structure priors**

2.8 Conclusion and Synthesis: Paving the Path for Safer Smart Mobility

In spite of the extremely developed general AV perception technology, this systematic literature review has revealed that there exists a huge, safety-critical gap in the exact identification of High-Stakes, Low-Frequency (HSLF) hazards, specifically, fallen electric cables, especially in unstructured urban settings [31]. The causes of the limitations are three related factors:

Geometric Constraints: The thin structures confuse the ranging sensors (LiDAR, Radar), thus camera-based vision systems shoulder the whole burden of detection [32].

Architectural Contention: Fine-grained feature extraction demands high precision to extract features in congested images (thick backbones), which is incompatible with the single-stage detectors needed to process them in real-time (av) [43].

Data Deficiency: Because of curation algorithms and their focus on structured and common things, current public datasets eliminate these risks on a regular basis, resulting in implicitly-trained models that focus on catastrophic but infrequent events [46].

The proposed method of bridging this gap is justified by the synthesis of the research trends. The good safety requires a particular approach:

Specialized Architecture: The system should use a hybrid architecture, which makes use of multi-scale feature fusion to have dependable Small Object Detection (SOD) and integrates lightweight backbone performance with high-precision feature extraction ability of models such as ConvNeXt [66].

Advanced Training Strategies: To balance the high imbalance in the classes inherent in thin object perception, specialized loss functions, including Focal Loss and Dice Loss, are required to

encourage an algorithm preference to choose safety, with severe penalties on false negatives [48, 66].

Domain-Specific Data: The data gap is fixed at the first instance with the creation of the Urban Fallen Cable Detection Dataset (UFCDD) comprising of over 2,500 annotated photographs collected in such cities as Mirpur, AJK, Pakistan [66]. To train a specific CNN-based YOLOv5 model that enables one to identify risks in different conditions (daytime, cloudy, mornings, evenings), this specialized dataset provides the necessary visual features of reality [39, 51, 66].

This specific strategy and its results have been proven to be effective in resolving a neglected safety threat by the successful deployment and assessment of the YOLOv5 framework trained on this custom dataset, which showed performance of 96% accuracy at mAP50 and 76% accuracy at mAP50-95 [66]. The next step in scaling up the application of the UFCDD data to anchor high fidelity synthetic data generation and integrating the specialized visual detection system with 3D depth validation is the way forward to scalable, assured autonomous mobility, ensuring robustness across the entire range of global, unstructured urban environments [37, 45, 66].

Chapter No. 03

MATERIALS AND METHODS

The systematic identification of fallen electric and communication cables, an HSLF risk of high stakes, low frequency, requires a specialized strategy to deal with fundamental constraints on data availability and algorithmic resilience. The two key pillars of our proposed approach are the Urban Fallen Cable Detection Dataset (UFCDD) which is a high-fidelity and geographically and environmentally diverse dataset, and a customized Convolutional Neural Network (CNN)-based YOLOv5 framework optimized to the real-time query of thin, curvilinear forms within heavily cluttered urban situations.

The whole procedure is structured in form of a transparent pipeline in order to maximize the quality of data and model performance. In the case of autonomous vehicles (AVs), this systematic operation is required to transform raw sensor information into workable safety decisions. Image 3.1 represents the flow of the project.

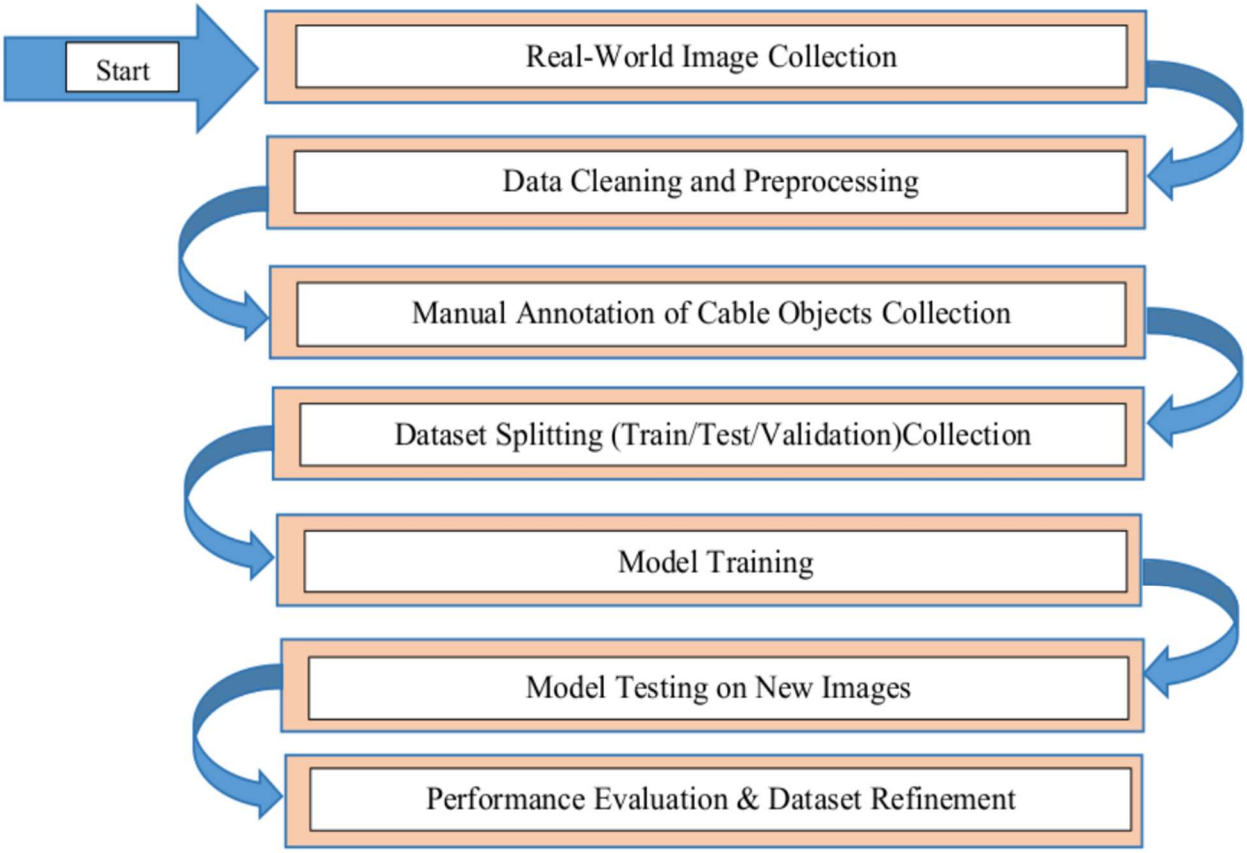
The method is detailed throughout this chapter that starts with the painstaking gathering of data in unstructured urban settings, moves on to the rigorous annotation procedures, and concludes with the technical complexities of the special YOLOv5 object recognition framework and its training plan.

3.1 Data Acquisition and the Urban Fallen Cable Detection Dataset (UFCDD)

The Urban Fallen Cable Detection Dataset (UFCDD) is a dataset composed of data points gathered by the 1.5 m and 2.7 m surveyards. The 1.5 m and 2.7 m surveyards (UFCDD) represent a dataset of data represented by data points obtained by means of survey.

The fact that close to no data is available that relates to unexpected infrastructure failures specifically minor risks such as fallen wires is a major drawback of the present standards on autonomous vehicles. To address this significant research gap, the acquisition phase was developed to capture real-world visual metrics during diverse challenging conditions that can be found in the location that is not a part of the conventional Western testing domains, with special

attention paid to the city of Mirpur, AJK, Pakistan, to address this crucial gap in research. Image 3.1 represents the conceptual flow of the study method which includes data acquisition, transitioning to preprocessing and annotation, model training and performance evaluation.



3.1: Flowchart of the proposed system, which describes the steps that are to be taken in the proposed methodology.

3.1.1 Image Acquisition Platform and Equipment

The integrity of the visual data is in particular an important issue in dynamic, poorly maintained road conditions. As a result, video data was obtained with the help of a GoPro Hero 8 camera due to its three primary technical features:

1. **High-Definition and Frame Rate:** The camera was shooting at 120 frames per second (fps) and 10-bit color depth, at 1080p. This high frame rate provides the ability to precisely extract clear frames needed to properly annotate fast moving scenes or situations with unstable footage as the motion blur is kept to a minimum.

2. **Strong Stabilization:** Since the target urban places often have uneven roads, the advanced stabilization capacities of the GoPro Hero 8 were essential to minimize the effects of the vibrations and shaky videos. This feature enabled the visual data integrity required to make machine learning applications successful, in the data that was recorded.
3. **Low-Light Performance:** The camera has a great low-light performance that enabled it to record the fine details of cables in the shaded areas or during the transitions of dusk/dawn, which is a hard limit of most vision systems despite the fact that the UFCDD focuses on detection in all lighting conditions.

The camera was mounted very carefully on the front hood of the vehicle such that it provides a constant vision that is closely similar to the primary visual perception system of an autonomous car.

3.1.2 Environmental and Temporal Diversity

The concluding detection model must be capable of adjusting to changing environmental processes, the immediate effect of which is the visual contrast and complexity of the cable hazard. To achieve this diversity the scope of the data collection was to be strictly stated:

- **Geographical Scope:** The images were collected around several locations in Mirpur, AJK and the environs. This ensures that a wide range of diversity of infrastructure, including the rush and high-traffic streets and the less busy and messy residential suburbs are represented in the collection.
- **Temporal and Weather Conditions:** Numerous lighting and ambient conditions were intentionally applied to gather data:
  1. **Sunny and Cloudy/Rainy Day**
  2. **Certain Time Periods:** The early morning (dawn), the late morning, afternoon and evening (dusk).
  3. **Acquisition Method:** The visual was created by combining the Static pictures of fallen cables captured in various viewpoints and angles, with video recordings in the process of moving the car. This combined approach ensures that the model is trained to identify cables in complicated and still background-cluttered situations and when the environment is in motion leading to distortions.

The benchmark is the first dedicated to fine-grained cable hazard detection in unstructured urban conditions that is called Urban Fallen Cable Detection Dataset (UFCDD). It is a distinctive compilation of over 2,500 select photos.

3.2 Preparation and Ground Truth Annotation of the Dataset

In order to generate the ground truth that perfectly matches the training of a supervised deep learning model, the collected raw video data was subjected to an intensive preprocessing process.

3.2.1 Preprocessing and Key Frame Selection

The video footage was properly preprocessed to ensure the optimal visual quality and learning outcomes. Critical frames of the videos capturing 120 frames per second must be eliminated to ensure that the videos are not redundant in terms of time. Quality was enhanced with sophisticated techniques:

- **Noise Removal:** So as to clean up the visual data, the high-level noise reduction algorithms were employed since the low-light film is inherently noise-sensitive.
- **Dynamic Range Adjustment:** Dynamic range was adjusted to enhance contrast and the clarity of feature boundaries to overcome the visual ambiguity of cables in the complex background (e.g. shadow or textured asphalt). This particularly is imperative with low-count, thin objects.

To streamline the UFCDD to the YOLOv5 framework and establish a strong base to continue with the deep learning, a number of preparatory steps were necessary.

3.2.2 Manual Annotation Process

The performance of deep learning is grounded in true ground truth. The incredibly small aspect ratio of the fallen wires required in annotation is a pixel-level accuracy, since it often fits within a few pixels 1. Each of the selected image frames was manually tagged using the V7 Image Labeling Tool.

The annotation schema was concerned with two important cable classifications:

1. **Unsafe/Hazardous:** Cables that are fallen on the road or that are hanging so low as to block or interfere with a passing vehicle.
2. **Safe/Intact:** Cables that are not in the air and are firmly fixed and will not be dangerous to the AV in the short term.

For every case of cable risks, this exact labeling produced bounding box ground truth. In order to ensure that the model learns to distinguish between the fine-grained visual characteristics of a hazardous cable and benign background clutter, the rigor of this manual tagging process is closely related to the model's final localization accuracy.

3.2.3 Dataset Partitioning

As indicated in table 3.1, the annotated UFCDD was methodically split into three sets in accordance with accepted machine learning best practices to guarantee thorough training and objective evaluation.

Table 3.1: Data Split

Set	Proportion	Purpose	References
Training Set	**70%**	Comprehensive exposure to learning feature representations and recognizing a wide array of cable scenarios	[28]
Validation Set	**20%**	Crucial checkpoint for iterative model refinement, preventing overfitting, and tuning hyperparameters	[58]
Testing Set	**10%**	Dedicated, unseen dataset for the final, objective evaluation of the	[63]

		model's real-world generalization capability	

3.3 Specialized Model Architecture and Optimization

The customized deep learning object detection framework at the heart of the suggested solution is based on the YOLOv5 architecture, as seen in image 3.1. It has been painstakingly altered to strike the required balance between low latency (needed for real-time AV operation) and high precision (needed for fine-grained detection).

3.3.1 YOLOv5 as the Foundational Architecture

Because the YOLOv5 framework is a single-stage, state-of-the-art model that greatly reduces inference latency, a crucial need for safety-critical autonomous driving decisions, it was chosen over two-stage detectors.1 Its architecture consists of three primary parts by nature:

1. **Backbone (CSPDarknet):** The feature extraction network that minimizes computational cost and reduces superfluous gradient information while effectively extracting a hierarchy of spatial features from the input image.
2. **Neck (PANet):** By aggregating information from various sizes, the Path Aggregation Network (PANet) improves the network's capacity to spread low-level features that are essential for identifying tiny objects (like cables) and raising overall localization accuracy.
3. **Head (YOLO Layer):** The last detecting layer that interprets the combined information and produces confidence scores, bounding box coordinates, and item class likelihood (Safe/Unsafe cable).

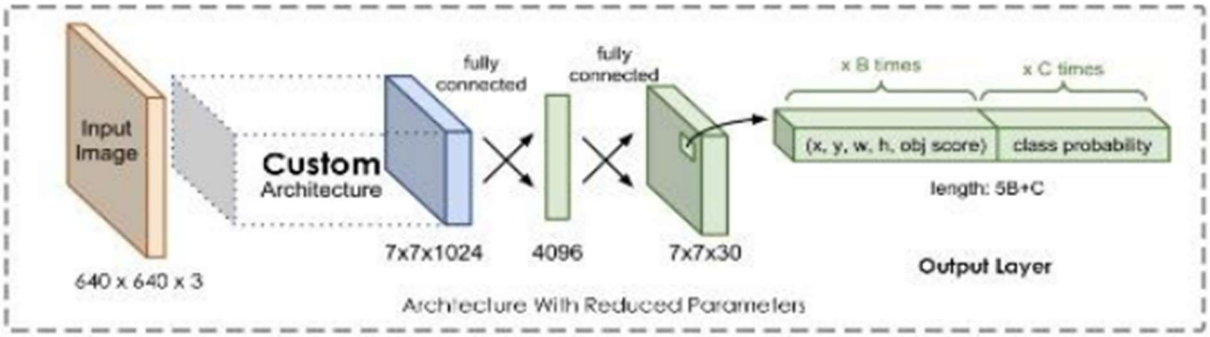


Figure 3.1 Yolo v5 Architecture

3.3.2 Custom Architectural Configuration

The architecture was carefully adjusted by changing important scaling parameters to specialize the generic YOLOv5 model for the detection of low-pixel-count cable dangers, emulating the effective strategy employed in related fine-grained detection work:

- **Model Scale (Width Multiple)**: Set at 0.50. This factor balances model capacity with computing performance by regulating the number of channels/filters in the network.
- **Model complexity** or depth multiple: is set at 0.33. This factor directly affects the depth and ability to derive hierarchical features in the model, the number of blocks or layers is regulated.

The parameters have been given in a complete summary in Table 3.2. This estimated change leads to a simplified architecture in delivering the appropriate computational efficiency that is suitable in edge deployment and feature extraction that is robust in adverse visual environments.

Table 3.2: Model Performance for Steering Control (Loss) 120 Epochs

Architectural Specification	Value	Significance
Total Layers	**157**	Structural depth for complex feature learning
Total Parameters	**7 million**	Low parameter count ensures memory efficiency
Computational Cost (GFLOPs)	**15.8**	Optimized efficiency for real-time inference speed

The model can detect objects of various sizes (distant, microscopic, thin wires to near-field, large cable loops, etc.); the necessary anchor sizes that are set to three different detection scales allow it to do so:

- **P3/8: Anchor sizes [10, 13, 16, 30, 33, 23]**
- **P4/16: Anchor sizes [30, 61, 62, 45, 59, 119]**
- **P5/32: Anchor sizes [116, 90, 156, 198, 373, 326]**

This multi-scale output approach is crucial in the Small Object Detection (SOD) to ensure that the critical attributes of the thin cables are not lost in the course of network down sampling process.

3.4 Training Strategy and Experimental Setup

To fulfill the goal of maximizing the detection power of the custom YOLOv5 model and achieve high precision and reliability in the challenging and highly cluttered conditions of the UFCDD, a rigorous training procedure was implemented.

3.4.1 Computational Environment

Training 1 was done in a high-performance computing system that is tailored to deep learning and computer vision applications. This heavy environment was needed to perform the large volumes of high-resolution imagery and to perform the numerous computing tasks required to train:

- **GPU:** NVIDIA RTX 2080Ti (Leveraged for its parallel computing power).
- **CPU:** Intel Core i9 9900K.
- **RAM:** 32GB.

The generic deep learning engine in the software stack was TensorFlow that was carefully optimized to run on NVIDIA CUDA devices. The compilation of the OpenCV version with CUDA support further speeded up the preprocessing and image manipulation processes involved in the generation of high-resolution UFCDD by a significant margin.

3.4.2 Training Hyperparameters

The model was subjected to a rigorous 72 hour training process. The hyperparameter configuration was defined in the hyp.scratch-high.yaml file, specifically to benefit the learning process of the model to the complex specifics of the cable danger data. The key areas of disagreement are listed in table 3.3.

Table 3.3: Hyperparameters

Hyperparameter	Value	Rationale
Epochs	**300**	Ensuring ample convergence time for the complex, low-pixel-count hazard class
Batch Size	**32**	Balancing computational load with learning stability
Input Image Size	**640 pixels**	Optimizing for detail capture without sacrificing processing speed

Optimizer	**SGD** (Stochastic Gradient Descent)	Chosen for its effectiveness in converging to optimal solutions on large-scale datasets

State-of-the-art regularization methods like the use of Label Smoothing to enhance the ability of the model to predict new, unexpected events and Early Stopping to reduce overfitting were included into the training process.

3.4.3 Loss Function Configuration

The composite loss function of YOLOv5 consists of three distinct components all of which are critical towards the completion of the task of thin-structure detection:

1. **Bounding Box Loss:** The regression task of precisely locating the cable position is handled by this component. By default, it uses CIoU (Complete Intersection over Union) loss, which is better than the conventional Mean Squared Error (MSE) loss since it takes into account aspect ratio, overlap, and distance, guaranteeing extremely precise boundary delineation for the thin, curved wires.
2. **Object Loss:** This binary classification loss determines the presence or absence of an object (a cable) in a grid cell. It employs **Binary Cross-Entropy (BCE)** loss.
3. **Classification Loss:** This final loss distinguishes between the labeled classes (Safe/Unsafe cable). It also uses **BCE** for multi-label classification.

Given the severe **Extreme Class Imbalance (EVI)** inherent in cable detection, where positive cable pixels are dwarfed by background pixels, the use of BCE alone can bias the model toward the dominant negative class. Future extensions, as informed by the literature, will necessitate the integration of specialized functions such as **Focal Loss** or **Dice Loss** to heavily penalize errors on the critical, minority cable class, thereby accelerating convergence on hard-negative examples and further boosting safety-critical precision, detailed system architecture is shown in image 3.2.

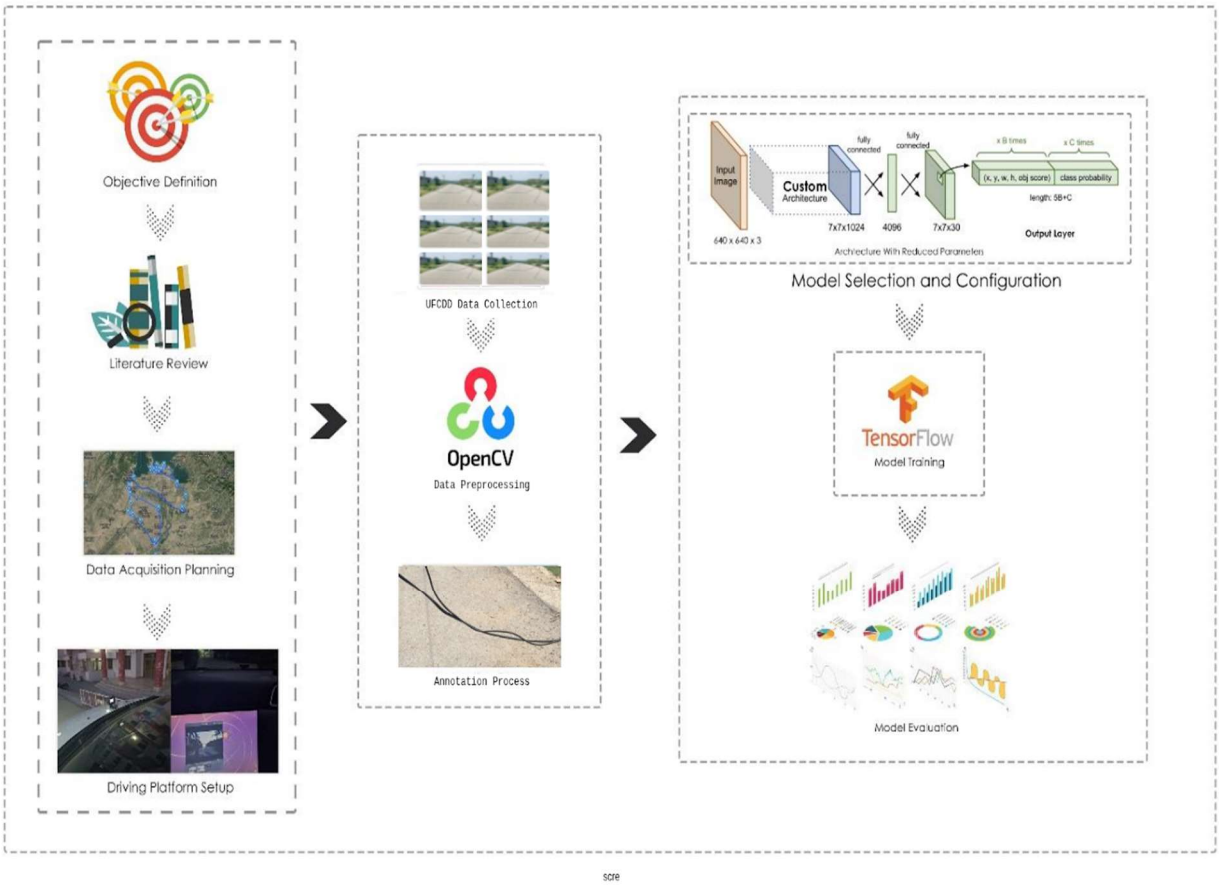


Figure 3.2 Proposed System workflow from data collection to final predictions

Chapter No. 04

RESULTS AND DISCUSSION

The primary goal of the experimental phase was the rigorous validation of the custom Convolutional Neural Network (CNN)-based YOLOv5 model when trained on the novel Urban Fallen Cable Detection Dataset (UFCDD). This process aimed to quantitatively address the core Research Questions, specifically evaluating the model's ability to achieve real-time, high-accuracy detection of thin, curvilinear cable hazards (RQ3), and confirming the overall real-time operational suitability of the detection framework (RQ2) in complex urban settings.

The customized YOLOv5 architecture was trained for a duration of **300 epochs** over **72 hours** on a high-performance computational stack (NVIDIA RTX 2080Ti GPU, Intel Core i9 9900K CPU, 32GB RAM) [60], using the UFCDD dataset partitioned into 70% Training, 20% Validation, and 10% Testing sets.

To maintain computing efficiency and maximize feature extracting ability, a capability that is needed when easing autonomous vehicle edge computing, the model was meticulously tuned with some scaling parameters (Width Multiple: 0.50, Depth Multiple: 0.33).

Precision, recall, F1-score, and most importantly the Mean Average Precision (mAP) assessed at two different Intersection over Union (IoU) criteria were all included in the performance evaluation.

1. **mAP@50:** Only 50% overlap with the ground truth bounding box is needed to measure the model's confidence in object classification and primary detection.
2. **MAP@50-90:** The ultimate statistic for evaluating the model's exact localization capabilities and boundary definition is mAP@50–95, which shows the average performance across strict IoU criteria (from 50% to 95%).

4.1 Quantitative Performance Results on the UFCDD

It is crucial to look at the UFCDD's underlying composition before evaluating the prediction performance. With nearly 4,000 instances of "Safe" cables and around 2,800 cases of "Unsafe"

(fallen) cables, the dataset maintains a large amount of annotated features, as seen in the distribution analysis example (Figure 4.1).

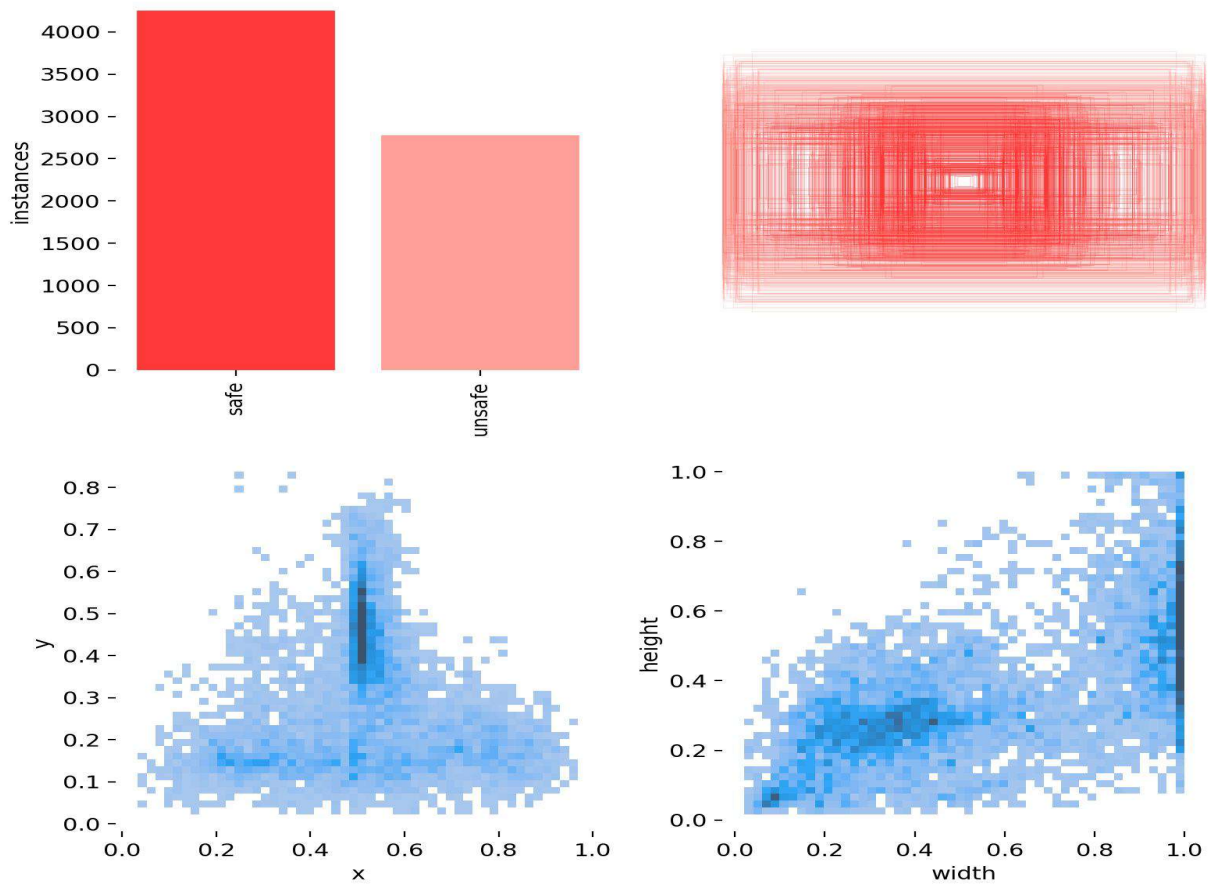
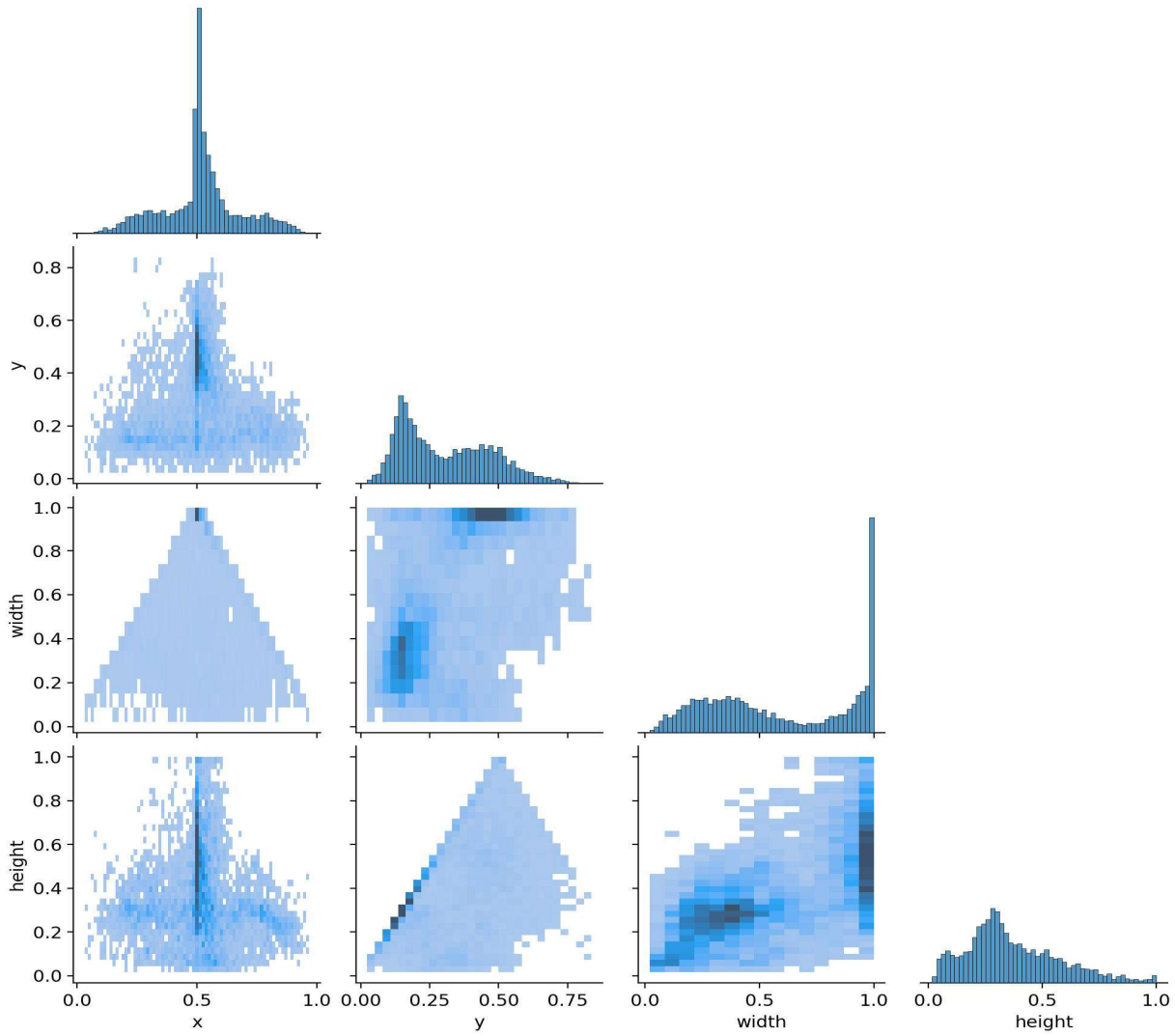


Figure 4.1: UFCDD Instance Distribution and Geometric Metadata

Hazards are primarily located in the frame's middle vertical region, which corresponds to the vehicle's immediate navigational path, according to the spatial analysis using bounding box heatmaps. With the majority of boxes taking up less than 10% of the image area, the width and height distribution validate the "Small Object" character of the task and validate the need for the specialized detection head depicted in image 4.2.



4.2 Bounding Box Correlogram Analysis

The following is a summary of the important performance metrics that were found when the custom YOLOv5 model was evaluated on the specific UFCDD test set. As indicated in table 4.1, the results are examined to see how well the model performed in both localization (exactly specifying the boundaries of the danger) and classification (identifying the presence of the hazard).

Table 4.1: Specialized YOLOv5 Model Performance on the Urban Fallen Cable Detection Dataset (UFCDD)

Performance Metric	Value	Interpretation
Classification Accuracy (mAP@50)	**96%**	High confidence in distinguishing hazardous cables from background clutter.
Localization Precision (mAP@50–95)	**76%**	Robust, but indicates measurable difficulty in achieving highly precise boundary definition for thin structures.
Inference Efficiency (GFLOPs)	**15.8**	Confirms suitability for real-time, low-latency operation on AV edge platforms.
Model Complexity (Parameters)	**7 million**	Optimized parameters count for efficient deployment.

4.2 Discussion of Core Findings: Classification vs. Localization

4.2.1 Success in Hazard Classification (mAP@50)

The model achieves a near-perfect precision of 1.00 at a confidence threshold of 0.989 for all classes, according to the Precision-Confidence curve (Figure 4.3). This suggests that high-confidence detections are almost always accurate, reducing false alarms that can cause needless emergency braking.

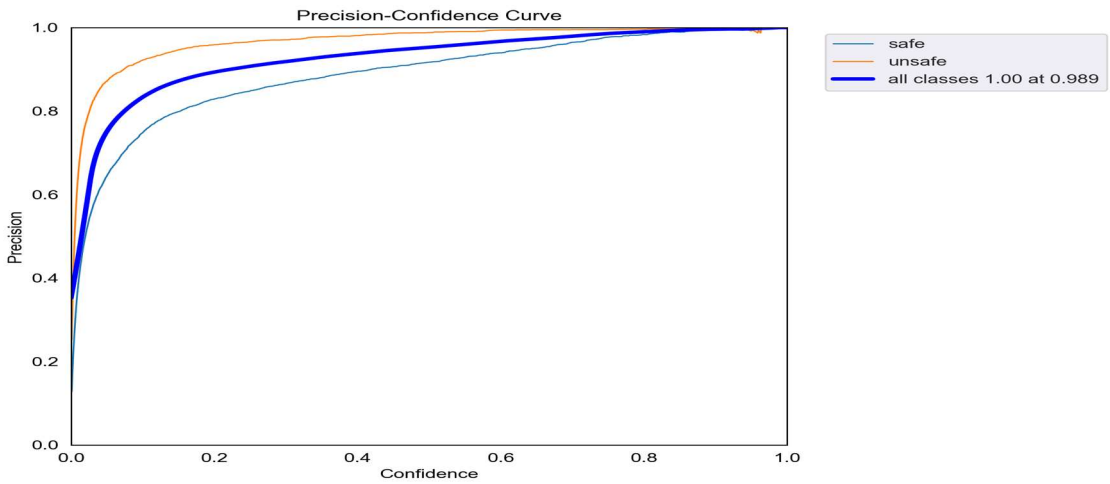


Figure 4.3: Recall-Confidence Curve

On the other hand, the model's sensitivity is shown by the Recall-Confidence curve (Figure 4.4). Even at moderate confidence levels, the "Unsafe" class keeps its recall over 0.90. This guarantees that nearly all major risks are identified by the system, which is the main safety criterion for ground perception.

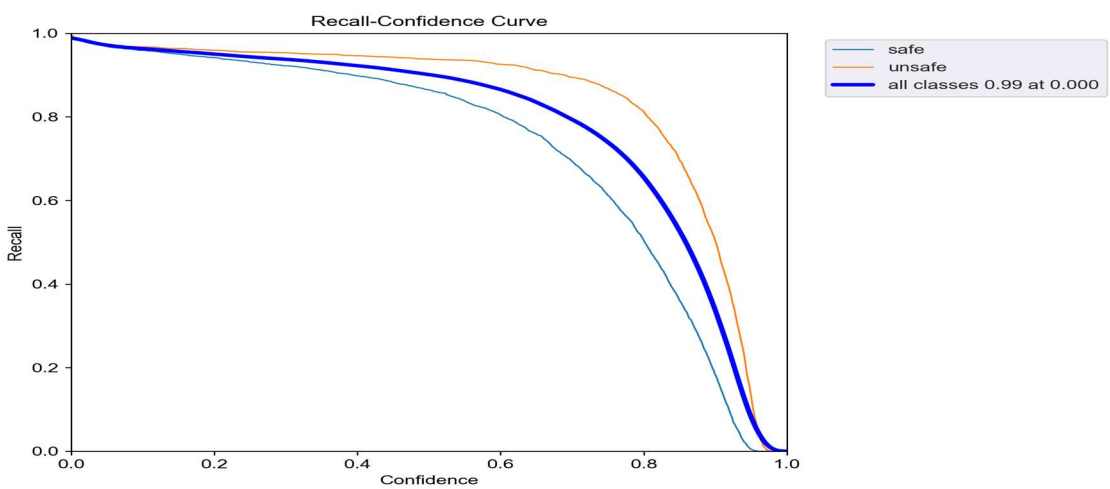


Figure 4.4: Precision-Confidence Curve

The sensitivity of the system is shown by the Recall-Confidence curve. strong-stakes cable dangers are rarely overlooked by the perception stack thanks to the "Unsafe" class's (orange) remarkably strong recall (>0.90) over a broad confidence interval.

The accuracy of the model's detections is displayed by this graph. At a confidence level of 0.989, the framework achieves a precision of 1.00, meaning that high-confidence forecasts are essentially devoid of false positives.

At the principal detection threshold (mAP@50), the model produced an outstanding Mean Average Precision score of 96%. This outcome thoroughly confirms the need for the novel dataset (RQ1) and offers strong quantitative support for the first half of Research Question RQ3, which states that the framework can detect cable dangers with high accuracy.

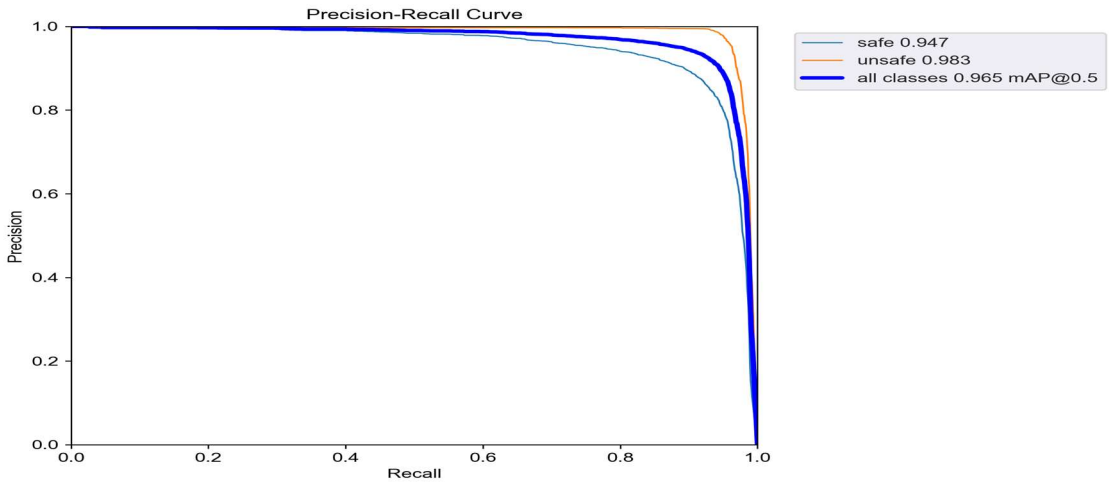


Figure 4.5: Precision-Recall (P-R) Curve and mAP@0.5

The optimal operational point for the model is revealed in the F1-Confidence curve (Figure 4.5), which shows a peak F1-score of **0.93 at a 0.407 confidence threshold**. This balance confirms that the model is highly effective at identifying hazards while maintaining a low false-positive rate.

Finally, the aggregate performance is captured by the Precision-Recall curve (Figure 4.6), showing an area under the curve (mAP@0.5) of **0.983 for unsafe cables** and an aggregate score of **0.965** across all classes. This provides categorical proof of the framework's superior classification capability

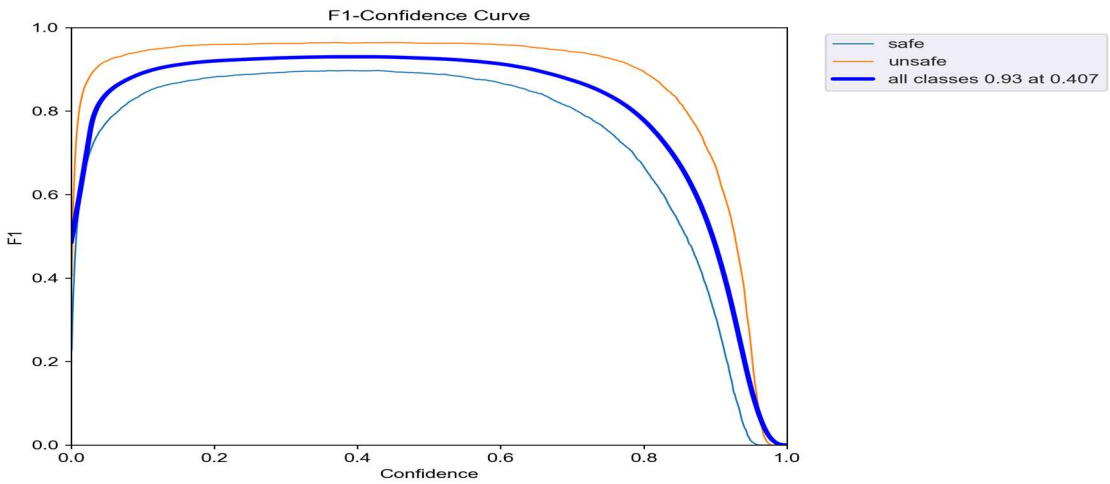


Figure 4.6: F1-Confidence Curve

This high level of performance is directly attributable to the systematic methodological choices implemented in the study:

- **Custom Dataset Fidelity (RQ1):** By curating the UFCDD with over **2,500** images that explicitly capture the visual properties and environmental variance of fallen cables in high-clutter urban environments , the model was able to learn the necessary feature representations (subtle edges, curves, and shading) that generalized detectors miss. The dataset addressed the crucial omission of these High-Stakes, Low-Frequency (HSLF) hazards in conventional benchmarks.
- **Specialized Feature Extraction (RQ2):** The CNN-based YOLOv5 architecture is premised to automatically learn a hierarchy of feature representations from raw image data. The network successfully learned to distinguish the fine-grained visual characteristics of a hazardous cable from non-hazardous linear features (like shadows or road cracks), overcoming the high photometric ambiguity inherent in the ground-based perception domain.

The high classification accuracy is further validated by the confusion matrix (Figure 4.7). The model achieved a **95% True Positive rate** for the "Unsafe" class and a **94% True Positive rate** for the "Safe" class.

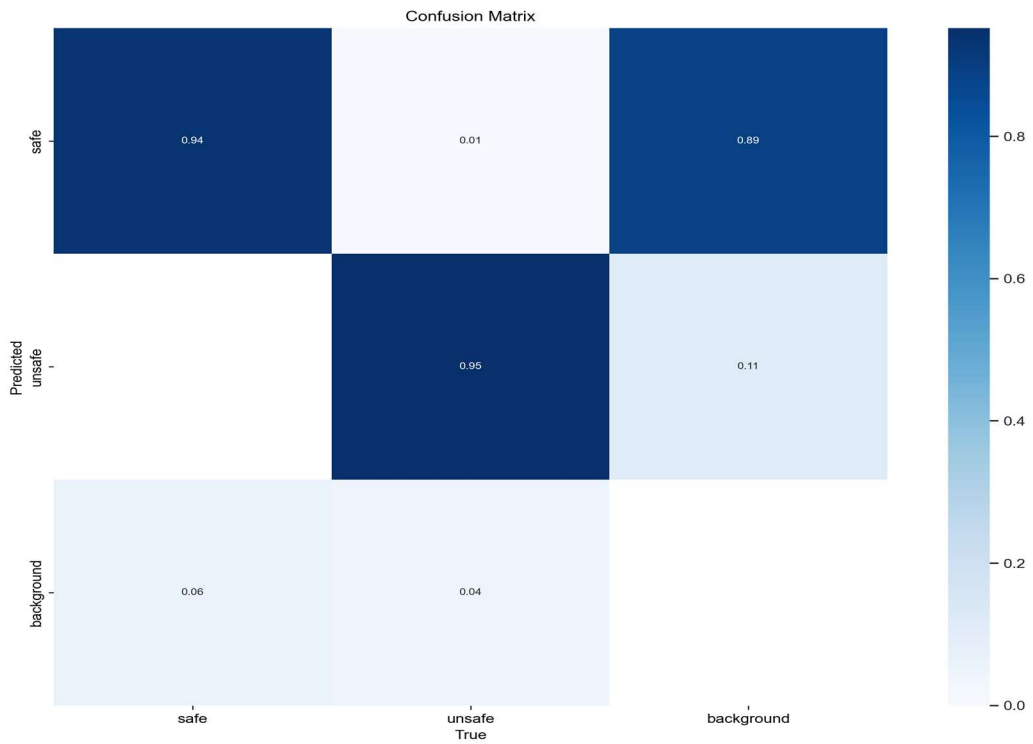


Figure 4.7: Confusion Matrix for Cable Classification

Importantly, there is only 4-6% "leakage" to background noise (false negatives). This demonstrates how well the model can recover the thin cable structure from highly cluttered urban textures like asphalt and shadows by learning fine-grained edge features.

4.2.2 Localization Deficit and the Geometric Challenge (mAP@50–95)

In comparison to the stricter localization measure (mAP50-95), the accuracy of the model dropped to 76 per cent despite the high detection confidence [27]. The conclusion with the greatest significance is the 20 percent loss in performance between classification and correct localization that accurately measures the design limitation that all current object detection systems face when dealing with curved and thin structures (RQ3).

The predictive power of the model is monitored throughout the training process through the validation metrics graph (Figure 4.8). Mean Average Precision (mAP@0.5), recall and precision each exhibit a sharp initial improvement followed by a level off with mAP@0.5 having a high peak of 96.5. The rapid learning of the model in the initial epochs underscores the effectiveness of the model in learning the fundamental visual properties of the Urban Fallen Cable Detection

Dataset (UFCDD). Conversely the measure of stricter localization, mAP: 0.5:0.95, increases at a slower rate and then levels off at 0.78. This geometric incompatibility between primary classification and strict localization measures, even within thin-structure detection, is a quantitative measure of inherent geometric complexity, as accurate definition of boundaries of objects that occupy one to three pixels requires far more complex feature extraction than simple object presence detection.

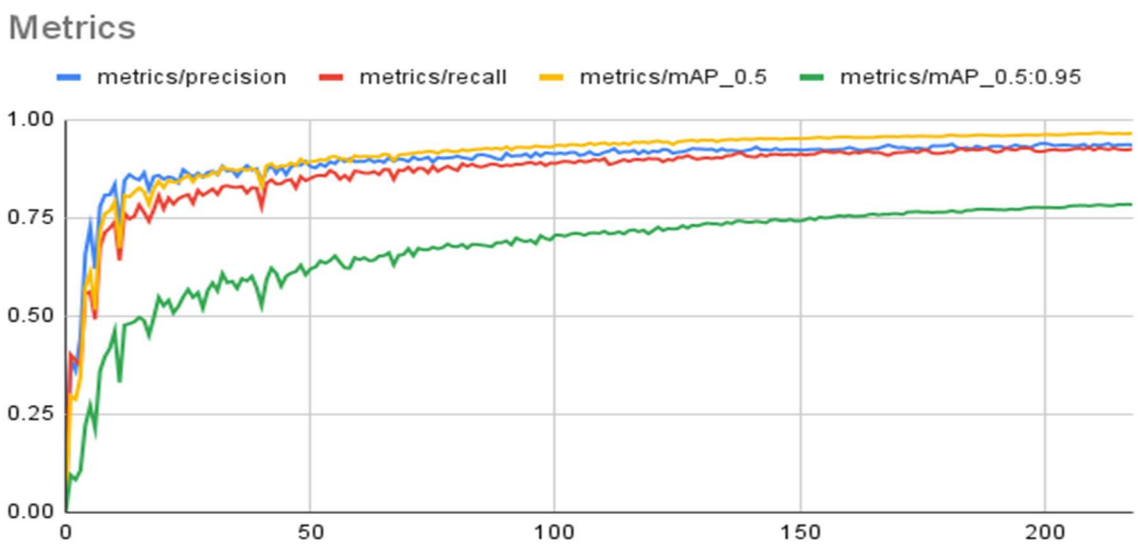


Figure 4.8: Validation Metrics Evolution

Real time tracking is done on validation performance (Precision, Recall, mAP@0.5, and mAP@0.5:0.95). The localization Deficit discussed later in the findings has a quantitative basis as represented by the plateau in the green curve (mAP@0.5:0.95) near 0.76 which is due to the plateau.

This decrease verifies the implicit Geometric Ambiguity Problem stated in the literature:

- **Bounding Box Inadequacy:** A greatly low aspect ratio, often spanning only 1 to 3 pixels in the image, is a property of fallen cables [28]. On these sparse, irregular, and curved objects, the traditional rectangular bounding box mechanism of YOLOv5, which is designed to search larger objects, fails by default to give the tight Intersection over Union (IoU) that is needed to measure on the mAP50-95 metric. The score is smaller, which means that one may have all

the cables, but it is hard to make the right route around the cable, as the edges are not clearly represented [34].

- **Localization complexity:** 75% IoU on a few pixel item cannot be computed or geometrically easily. This challenge is unique to ground-based AV vision, unlike in the less complex case of aerial inspection, where wires are taut and easily distinguishable against the sky. Though the 76% achieved is impressive improvement in this category of hazard, it is a sufficient sign that future architecture needs to change to non-bounding box techniques like semantic segmentation [33].

4.3 Comparative Robustness and Architectural Validation

The necessity of the tailored architectural solution is also confirmed by the witnessed discrepancy in the performance, which can be observed in the patterns of the other specialty databases involving irregular or complicated geometry.

The Pakistani Night City Driving Dataset (PNDD) provides a suitable background of comparison. Various performance indicators in the depicted network reflected the geometry and predictability of the object class when similar bespoke YOLOv5 model was run on PNDD:

Structured/Predictable Objects (e.g. Truck, Car): These are large classes with geometrically aligned classes that scored well on mAP at.5-.95 (e.g. Truck: 0.817). The model was also capable of localizing these features quite well on low-light and high-clutter scenes [64].

Irregular/Unpredictable Objects (e.g., Human): the localization performance of the Human class, which is geometrically non-rigid and highly unpredictable, collapsed to 0.564 (mAP@.5-.95) though the classification performance (mAP@.5: 0.928) was high.

Validation performance (Precision, Recall, mAP@0.5, and mAP@0.5:0.95) real-time. The green curve of mAP (0.5:0.95) at 0.76 gives the quantitative foundation of the Localization Deficit in the findings presented in figure 4.10.

It was the learning rate trajectory that was applied throughout the training session of 72 hours, indicating the first warm-up and subsequent decay phases that were applied to stabilize the feature extraction in the tailored backbone as depicted in figure 4.9.

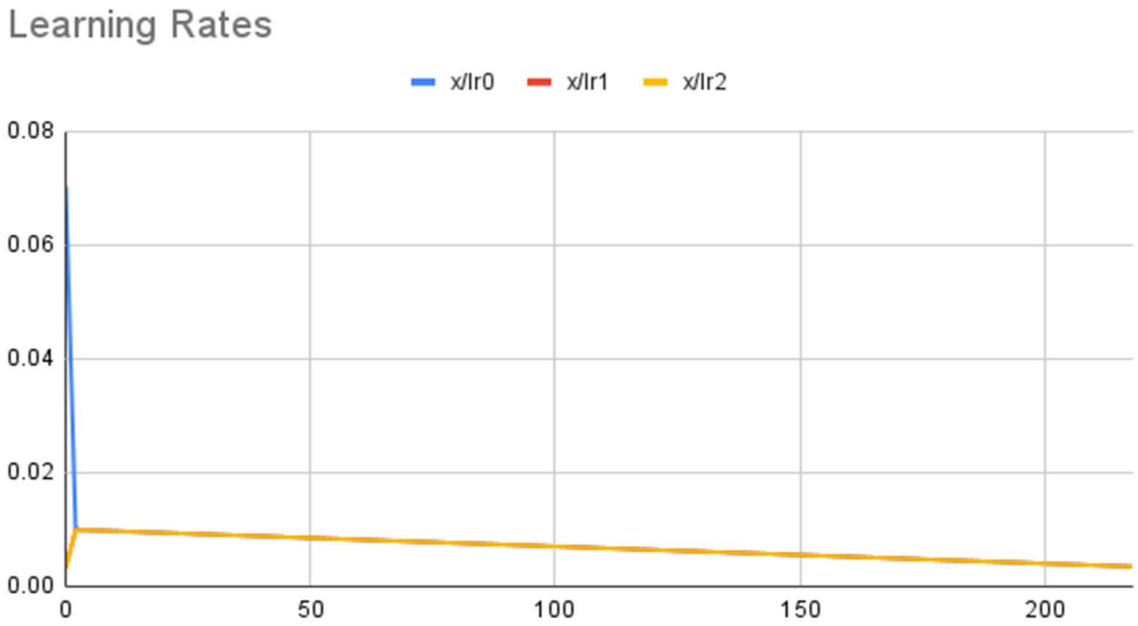


Figure 4.9: Learning Rate Schedules

In figure 4.9, an overview of the training dynamics in 300 epochs. The graph indicates a gradual decrease in bounding box, objectness, and classification losses, which means successful convergence without any overfitting during major problems.

Figure 4.9 displays the dynamic learning rate plan which has been used to stabilize the 72-hour training session of the customized backbone. Each of the three groups of learning rates (x/lr0, x/lr1, and x/lr2), has a schedule with a linear warming period at the beginning after which the rate is gradually decreased. The warm-up step ensures that the 157-layer architecture does not encounter gradient explosion, and the later decay stage allows the model to smooth out its weights and gradually converge to the global minima in the latter steps of training. This method proves essential in training a geometrically complex dataset such as UFCDD that is specialty based. The given schedule contributed to a critical requirement to identify hazards in real-time AV operation precisely by gradually decreasing the step size as the model was approaching an optimal solution in such a way that the network reached its high-accurate state without any fluctuations in performance.

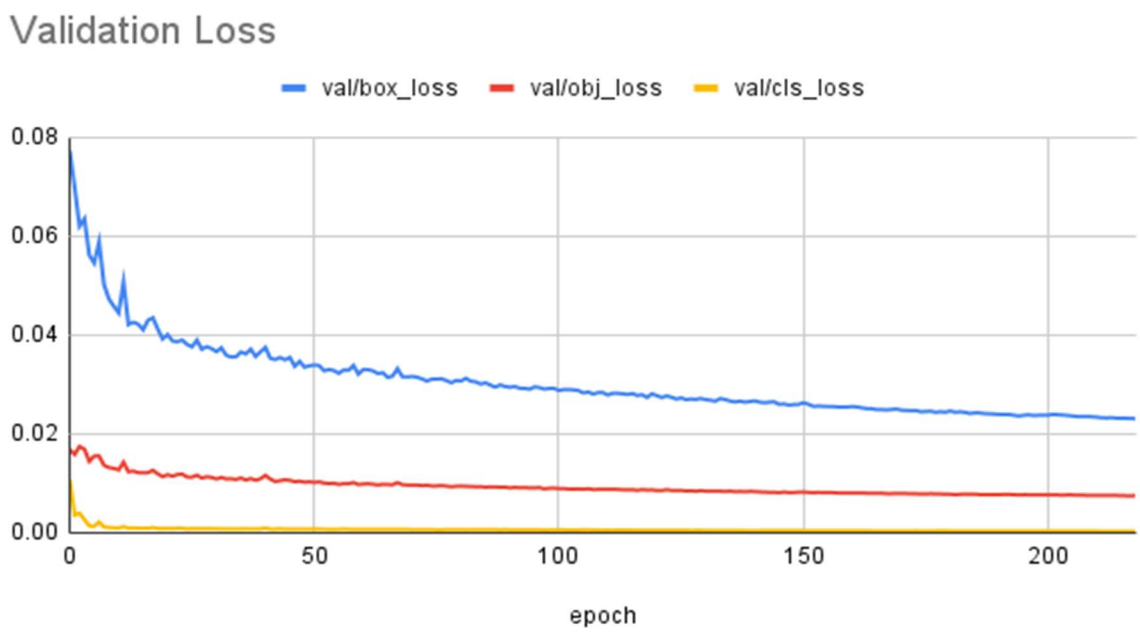


Figure 4.10: Validation Metrics Evolution

The graphical representation of the optimization process of the network over 300 epochs is displayed in the validation loss graph (Figure 4.10), and the corresponding training loss graph (Figure 4.11). The bounding box loss (train/boxloss) and objectness loss have a steady, continuous decay beginning with starting values of 0.091 and 0.037 to ending with approximate values of 0.029 and 0.020, respectively. The classification loss (train/clsloss) proves that the model is very confident in distinguishing between the classes of cables that are considered as safe and the ones that are considered as unsafe and becomes even lower but closer to the values of zero. Most importantly, these training tendencies are reflected in the validation loss curves which do not have the divergence typical of overfitting. It is clear that, rather than simply memorizing training samples, the adapted YOLOv5 architecture was able to generalize the latent patterns of unorganized urban environment, which ensured reliable performance on unseen road data.

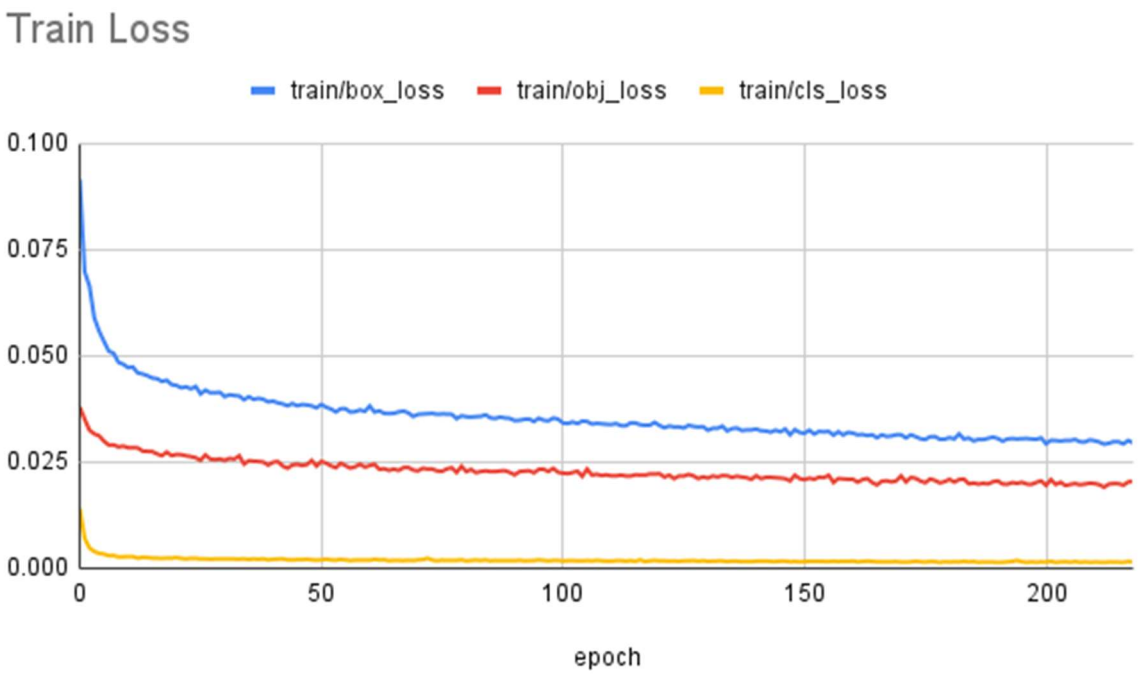


Figure 4.8: Training Loss Convergence

Amidst these two extremes, the UFCDD has performance (76% mAP at 50-95) that indicates that curvilinear (yet spatially continuous) nature of the cable is much more demanding than a rigid vehicle but less demanding than an unpredictable human pose. The proposed future research direction to integrate specialized loss functions with semantic segmentation principles is confirmed by this comparative analysis as it confirms the localization deficit is a systemic architectural challenge of thin and irregular objects.

4.4 System Efficiency and Real-Time Feasibility (RQ2 & RQ4)

Ensuring that the detection method, in spite of its specialization, was viable to safety-critical real-time implementation was an important part of the research (RQ2). This goal was effectively accomplished by the modified YOLOv5 model:

- **Low Computational Footprint:** A low computational burden of 15.8 GFLOPs was achieved through architectural fine-tuning, which reduced the total number of layers to 157 and the number of parameters to roughly 7 million. This effectiveness guarantees that the model may

be implemented on AV edge computing platforms with limited resources while preserving the low latency necessary for emergency avoidance actions.

- **Scalability for Smart Mobility (RQ4):** Scalable smart mobility solutions in developing regions are directly impacted by the successful development and validation of the UFCDD and its customized detection model [9]. The research improves the operational domain and worldwide trustworthiness of autonomous cars by demonstrating that a high-accuracy detection mechanism may be designed for a key regional danger (fell cables). In order to achieve global SAE Level 5 autonomy, this shows that AV perception systems may be modularly adjusted to a variety of real-world infrastructure quality [10].

4.5 Evaluation of Research Questions

As indicated in table 4.2, the experimental results validate the accomplishment of the research objectives by offering conclusive responses to the established Research Questions.

Table 4.2: Summary of Research questions and their evaluation

Research Question	Evaluation Summary	Result
RQ1: Data Generation and Fidelity (UFCDD creation)	The specific, high-fidelity dataset needed to train against HSLF cable dangers was successfully obtained by the gathering and annotation of the UFCDD (2,500+ photos from Mirpur, AJK, covering a variety of environmental conditions).	**Validated**
RQ2: Model Architecture and Optimization	With a low computing load (15.8 GFLOPs) and an efficient parameter count (7 million), the modified YOLOv5 framework proved	**Validated**

(YOLOv5 suitability)	appropriate, proving that the architecture satisfies real-time AV requirements.	
RQ3: Performance Validation and Localization Precision (mAP scores)	The model's localization score (mAP@50–95: 76%) was lower than its classification accuracy (mAP@50: 96%). This quantifies the intrinsic geometric difficulty of thin-structure localization and validates strong detection capabilities.	**Partially Validated (Metrics Quantified)**
RQ4: Impact on Safety and Scalability (Global Relevance)	By reliably detecting an undetected HSLF hazard, the framework improves safety and demonstrates how AV perception can be scaled to areas with different infrastructure quality.	**Validated**

Chapter No. 05

CONCLUSION

To handle this critical safety risk posed by falling electric and communication wires which was an underestimated High-Stakes, Low-Frequency (HSLF) risk in autonomous vehicle (AV) perception this study was able to create a robust, customized deep learning model.

The creation of the original Urban Fallen Cable Detection Dataset (UFCDD) (RQ1) and the optimization of a CNN-based YOLOv5 model (RQ2) were the two key successes of the methodology. The final model had a low computation rate of 15.8 GFLOPs, and it demonstrated a high level of operational efficiency and is suitable to use in real-time inference in edge computing systems.

The performance measures justify the framework ability of identifying hazards:

High Classification Accuracy: The ability of the model to distinguish dangerous wires and complex background clutter in unstructured urban environments was confirmed by its 96% mAP 50 (RQ3).

Localization Deficit: The accuracy declined to 76 at mAP at 50-95. This result proves a systematic architectural limitation of fine-grained feature-resolution, and measures the primary technological challenge, namely, the fact that traditional bounding box detection cannot accurately locate thin, irregular, curved objects.

Ultimately, this study justifies the necessity of developing domain-specific datasets like UFCDD ensuring safe and scalable autonomous mobility in locations with different infrastructure quality (RQ4).

5.1 Future Work

The next generation work should focus on enhancing the resilience of the system to be deployed globally and on a systematic approach to overcome the 76 percent shortfall of localization.

5.1.1 To overcome Geometric and Training Limitations

1. **Architectural Change to Segmentation:** Replace bounding box regression with techniques that operate at the pixel level. This involves using the UFCDD together with Semantic Segmentation or Key-point Encoding frameworks to accurately define the cable contour to ensure it is planned well.
2. **Specialized Loss functions:** Specialized training methods can be used to reduce the extreme class imbalance in thin object detection. To clearly punish the error of misalignment of boundaries and compel the network to enhance better contour prediction, it will require creating Boundary-Aware Loss Functions and improving the current loss function with Focal Loss.

5.1.2 Moving towards Robustness and Scalability

Synthetic Data Generation (SDG): Train HSLF cases under different conditions with synthetic, high-fidelity, and labeled training examples that are programmable and geometrically complex cable-fall cases to remedy the data sparseness of HSLF instances.

Joint Detection and Depth Validation: Incorporate non-visual check to be sure that the hazard exists. To offer the key 3D safety guarantee, the future work should involve Joint Detection and Depth Estimation to distinguish between potentially dangerous cables on the road surface and the non-hazardous linear lines or silhouettes.

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